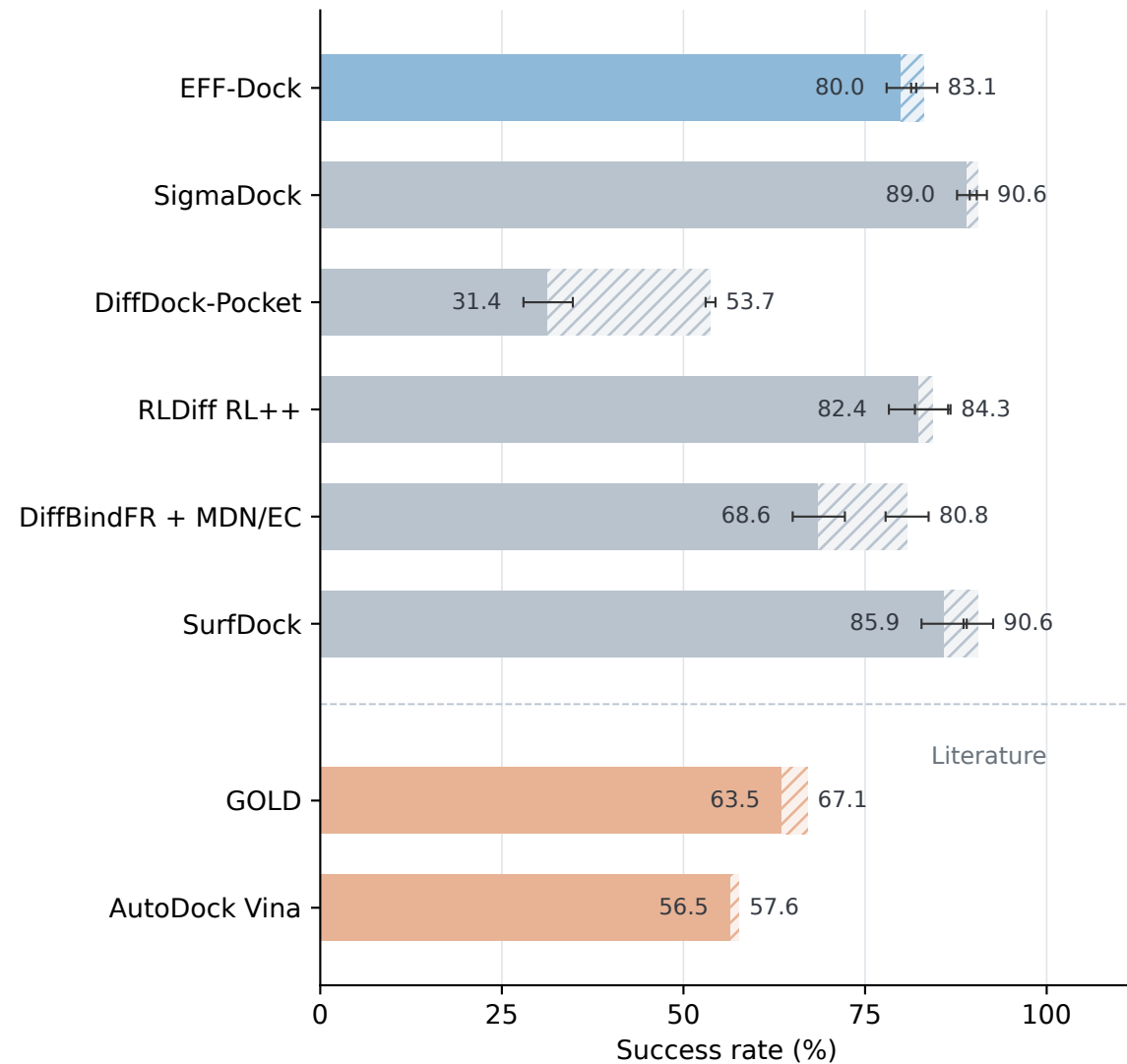
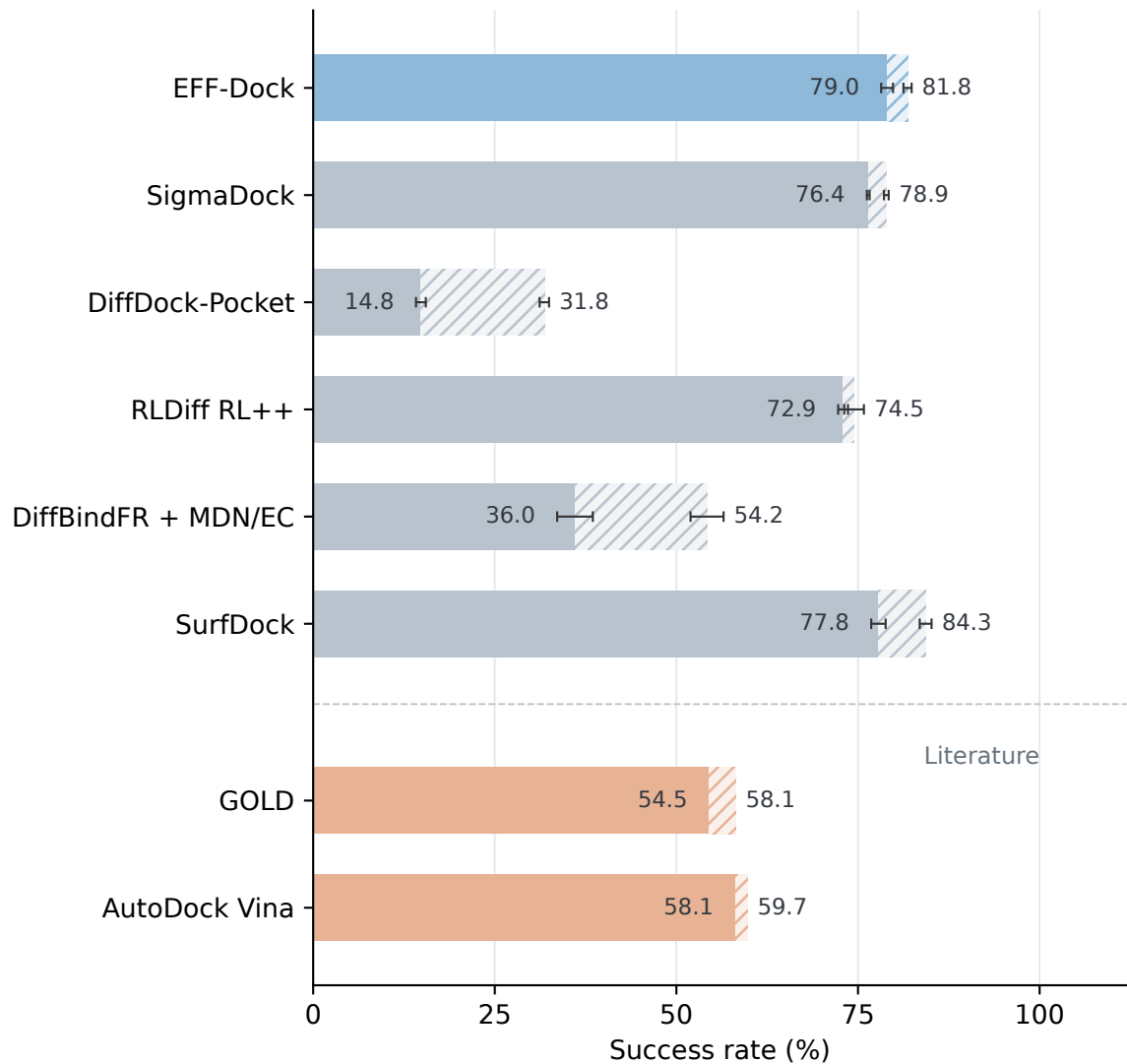
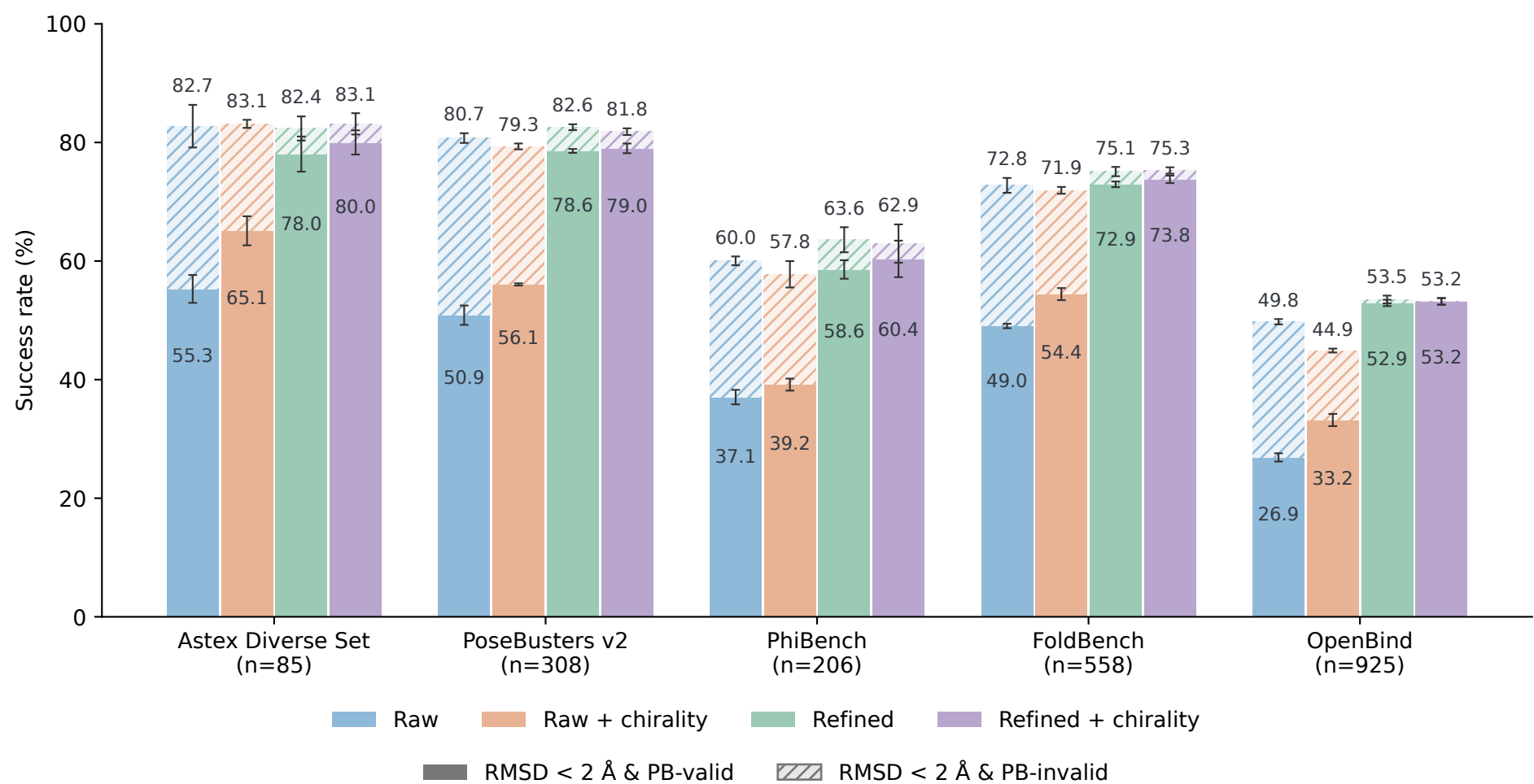
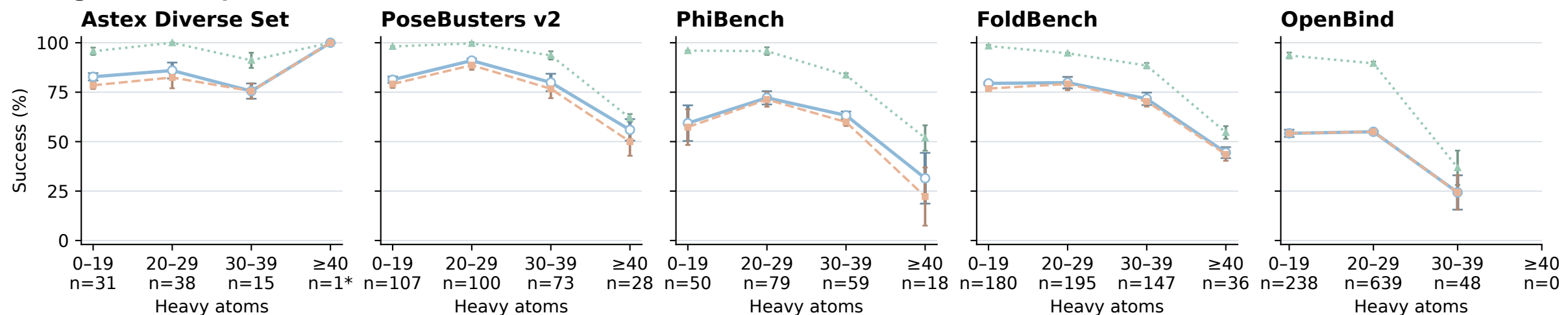


A Astex Diverse Set**B PoseBusters v2**

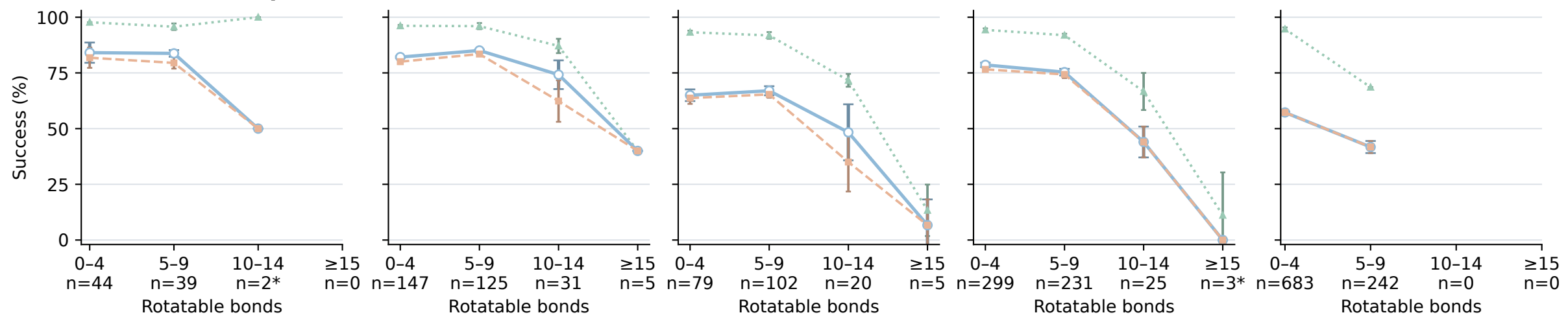
■ RMSD < 2 Å & PB-valid ▨ RMSD < 2 Å & PB-invalid



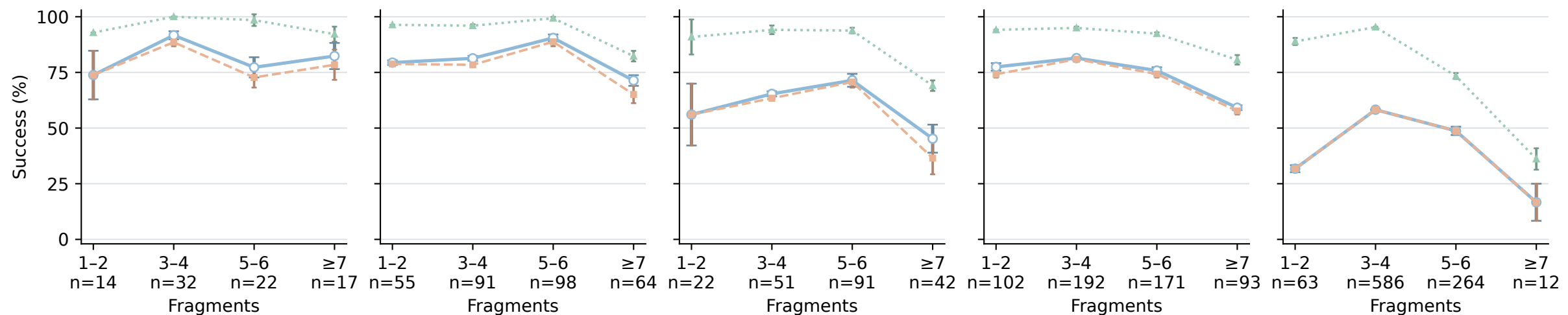
A Ligand size dependence



B Rotatable bond dependence

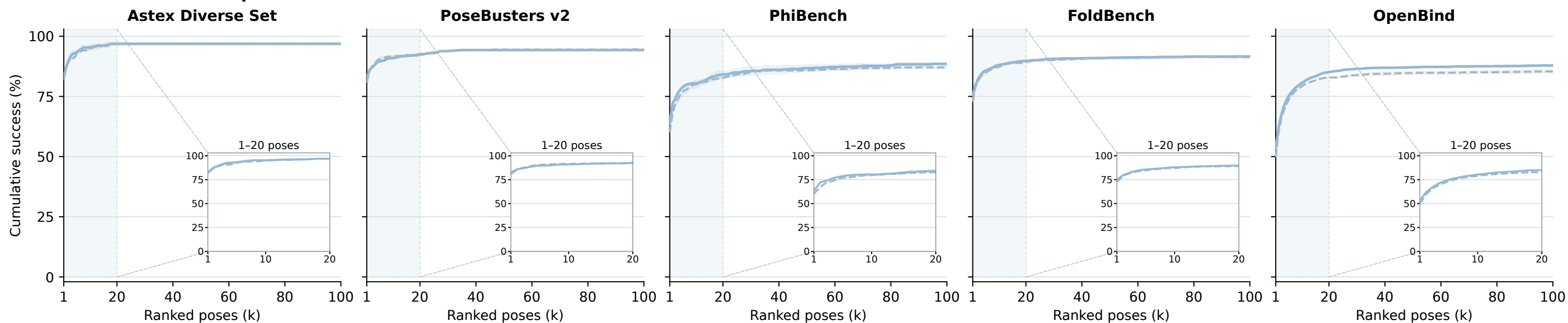


C Fragment count dependence

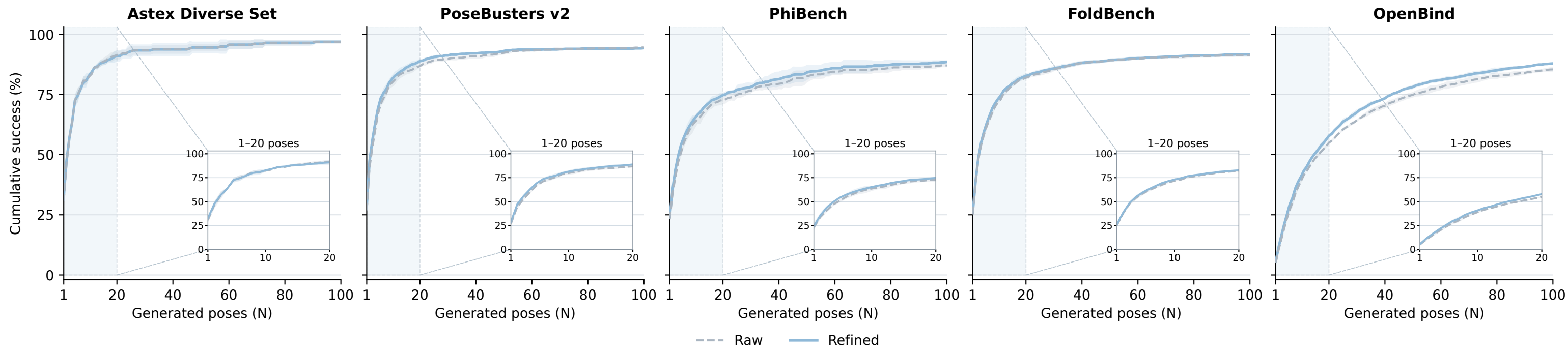


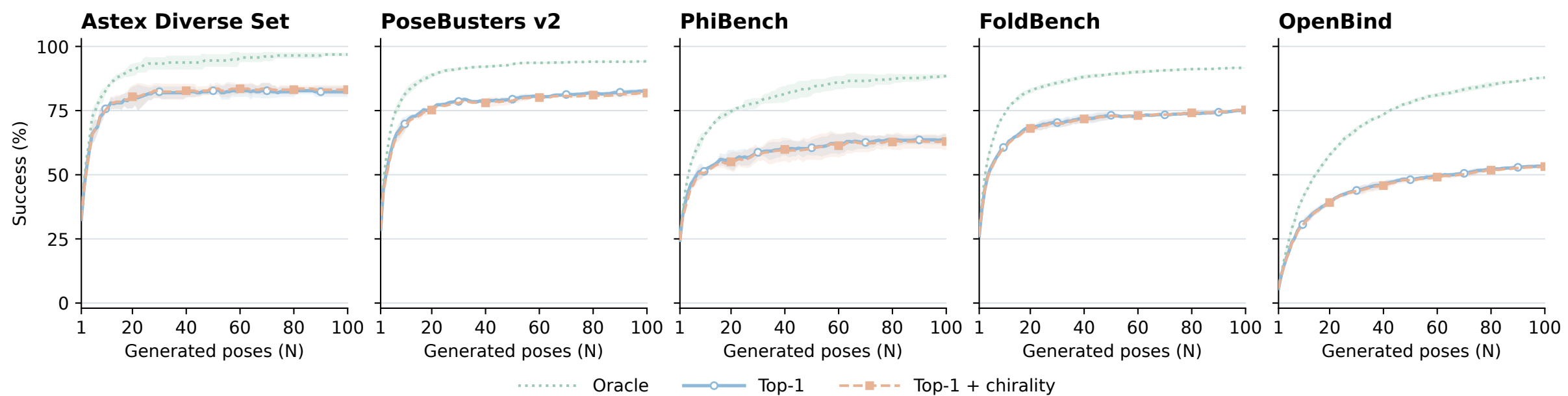
Oracle Top-1 Top-1 + PB-valid

A Confidence-ranked poses

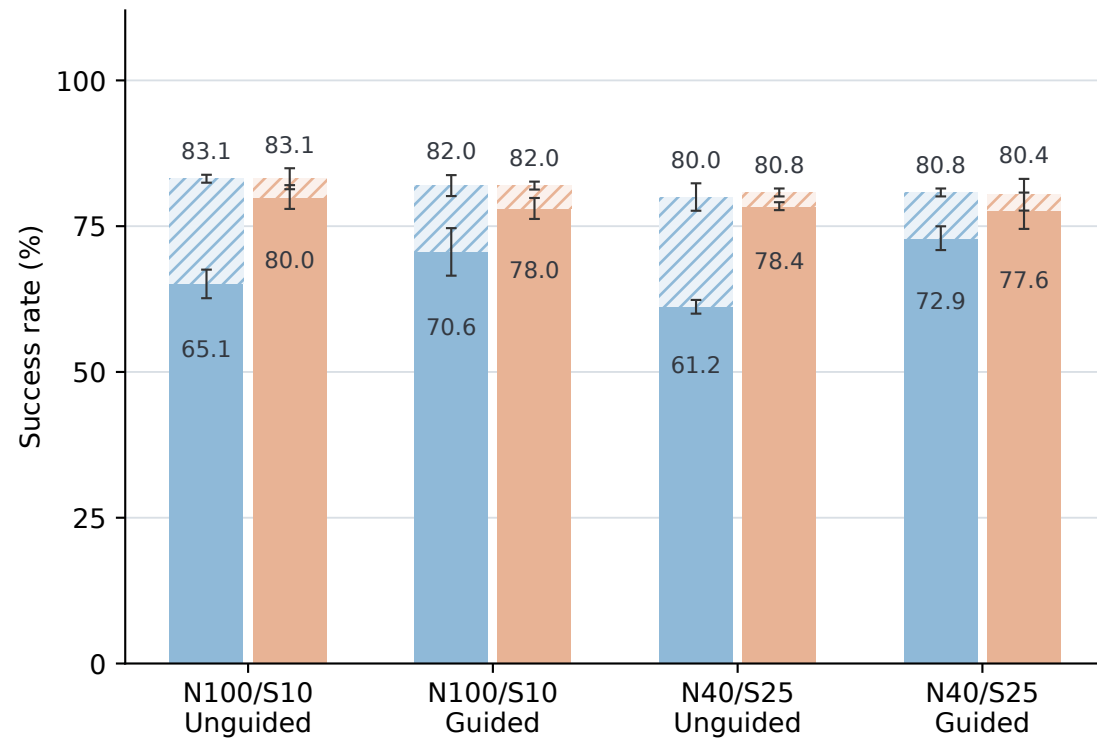


B Generation-order poses

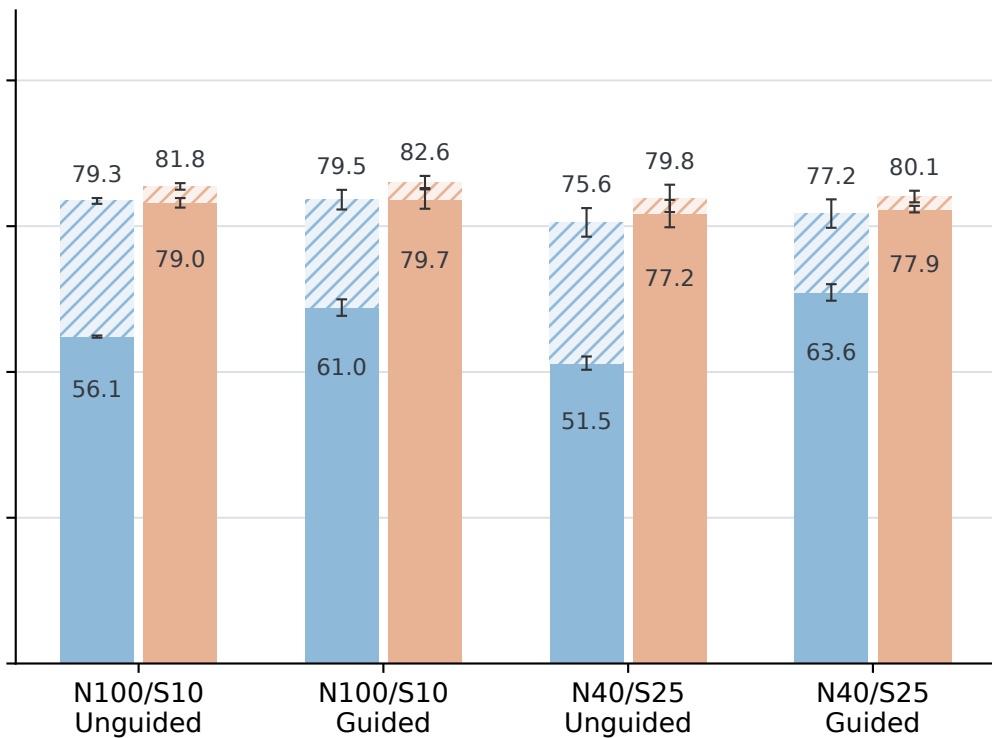




**A Astex Diverse Set
(n=85)**

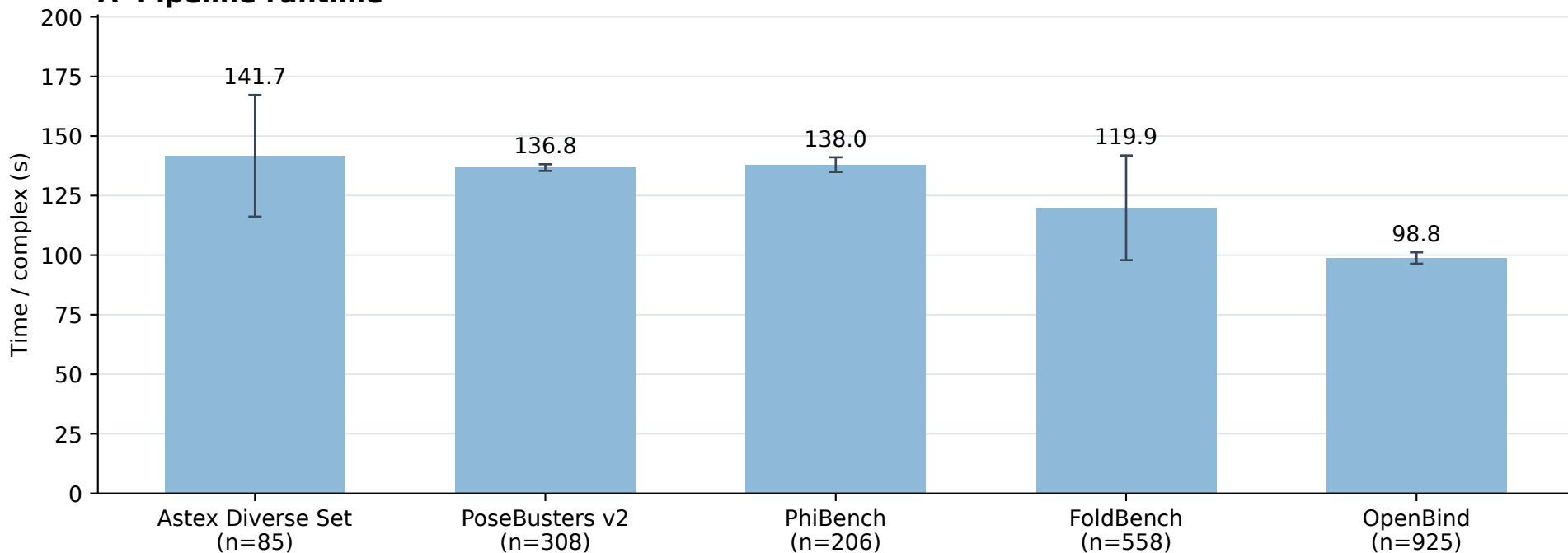


**B PoseBusters v2
(n=308)**

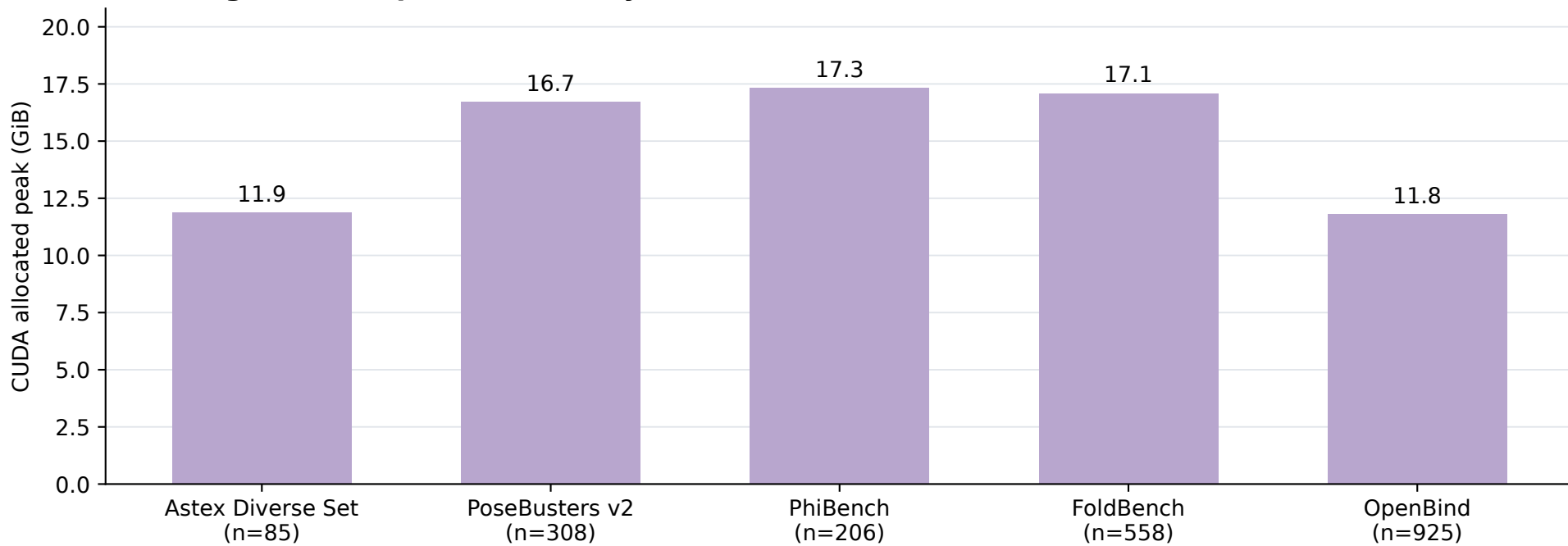


Raw Refined
RMSD < 2 Å & PB-valid
RMSD < 2 Å & PB-invalid

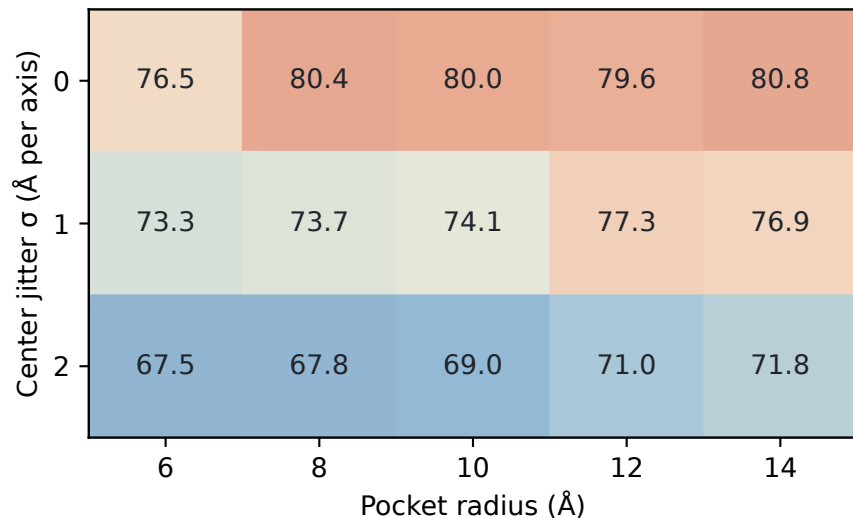
A Pipeline runtime



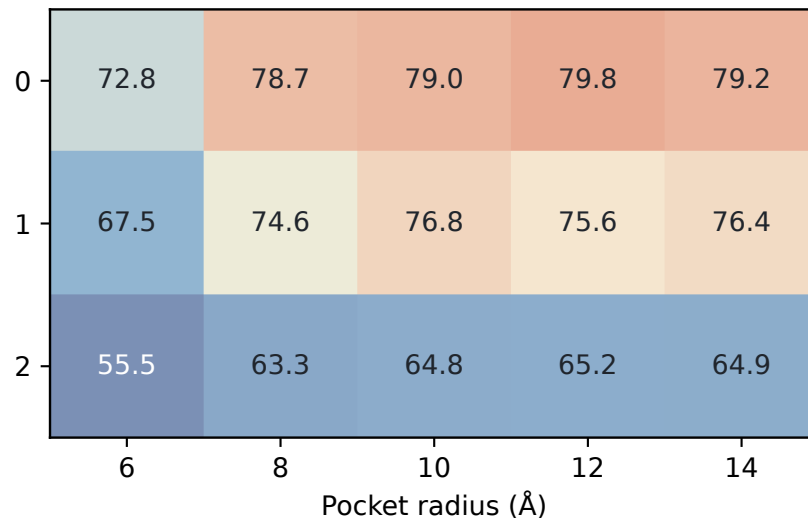
B Pose-generation process memory



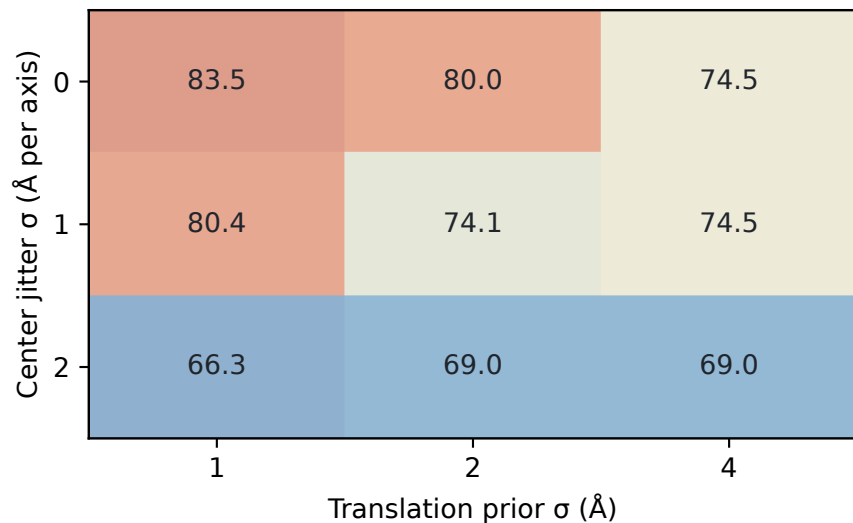
**A Astex Diverse Set
(n=85)**



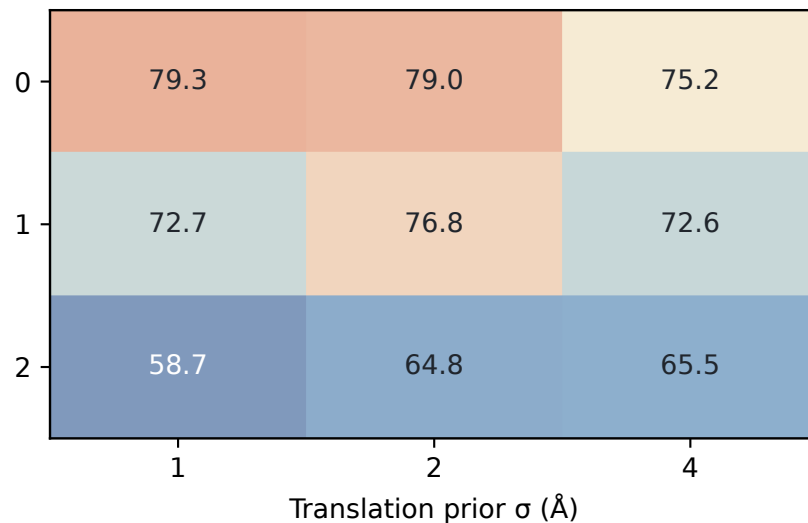
**B PoseBusters v2
(n=308)**



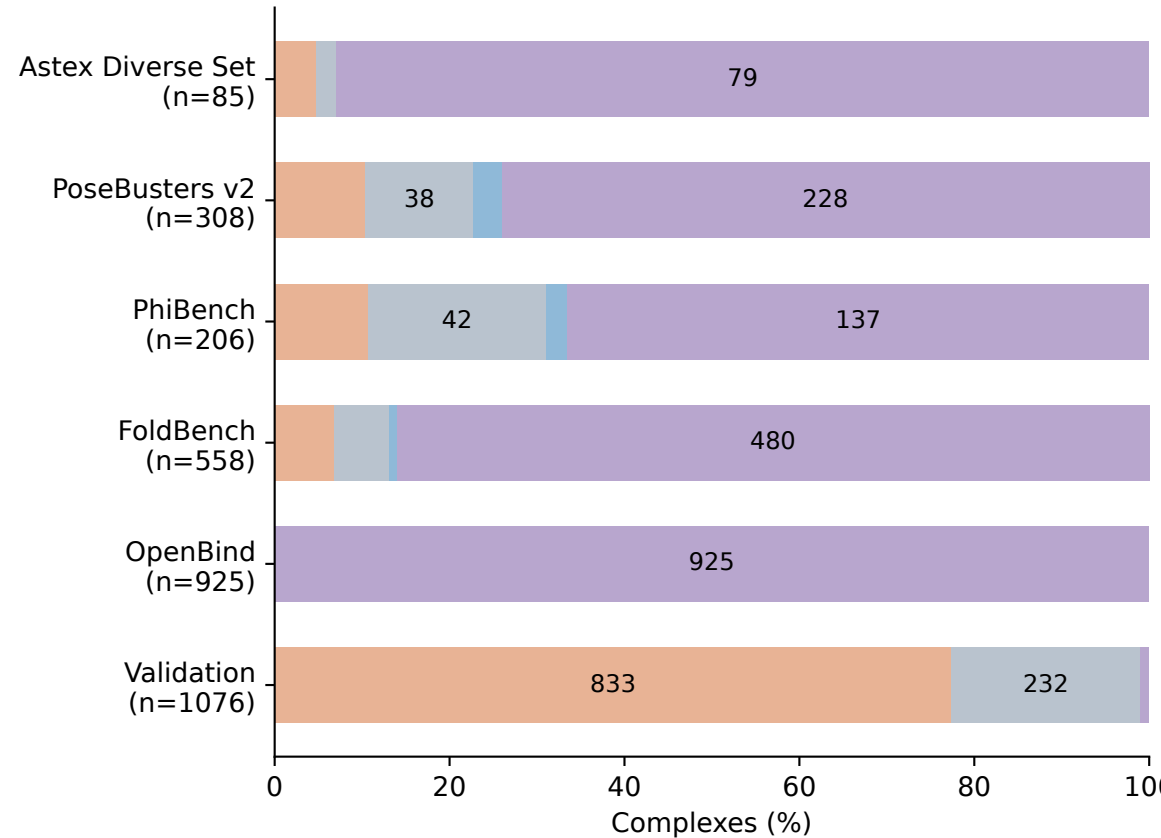
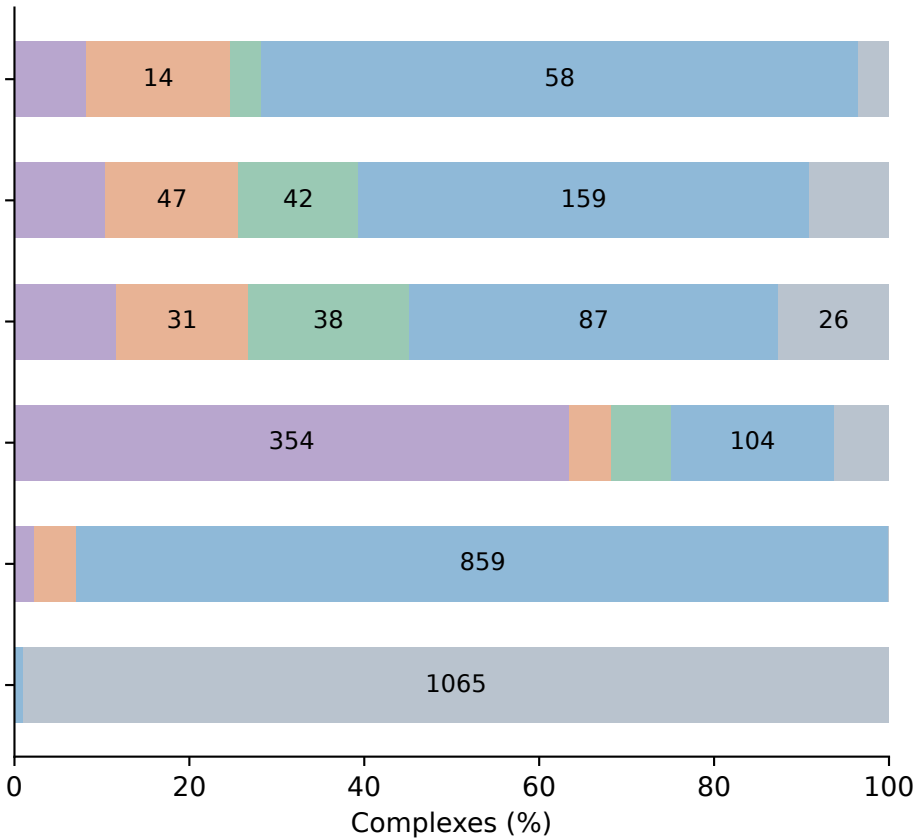
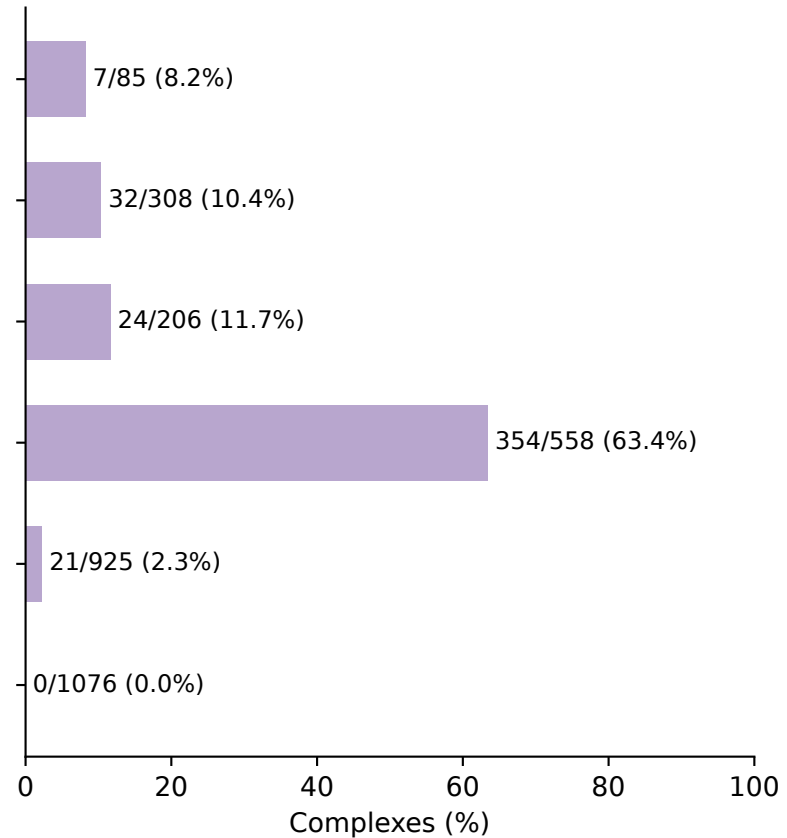
**C Astex Diverse Set
(n=85)**



**D PoseBusters v2
(n=308)**



RMSD < 2 Å & PB-valid (%)

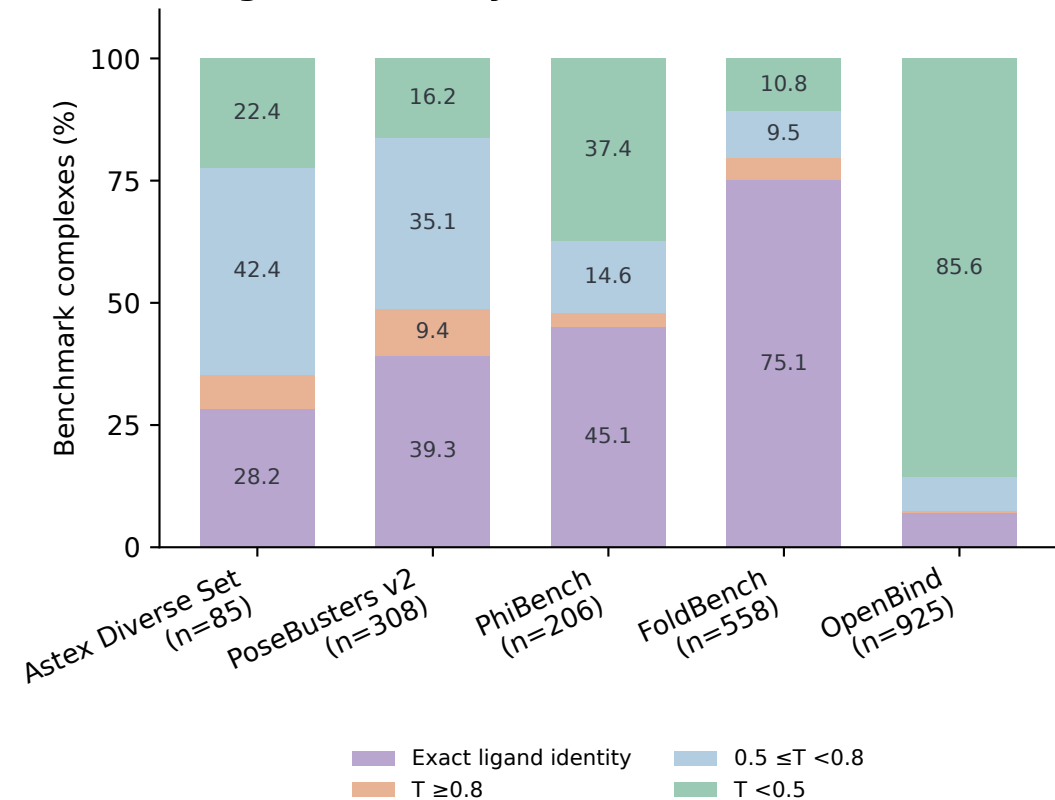
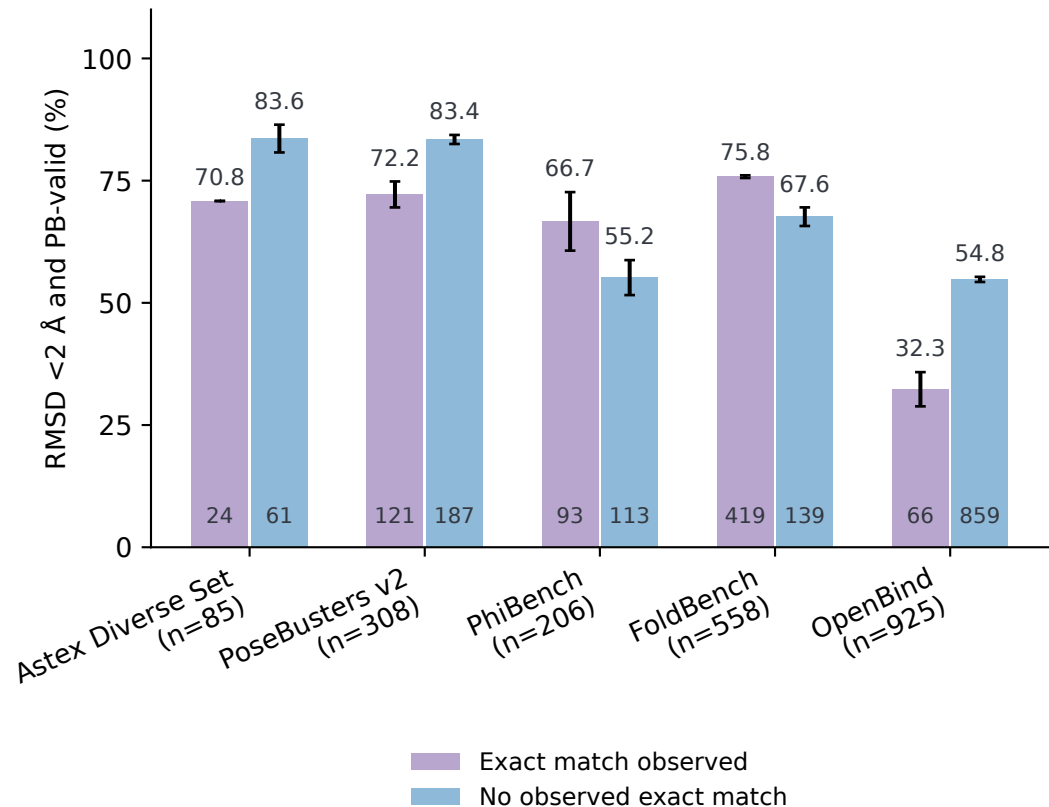
A Sequence identity**B Ligand-protein overlap****C Same training sample**

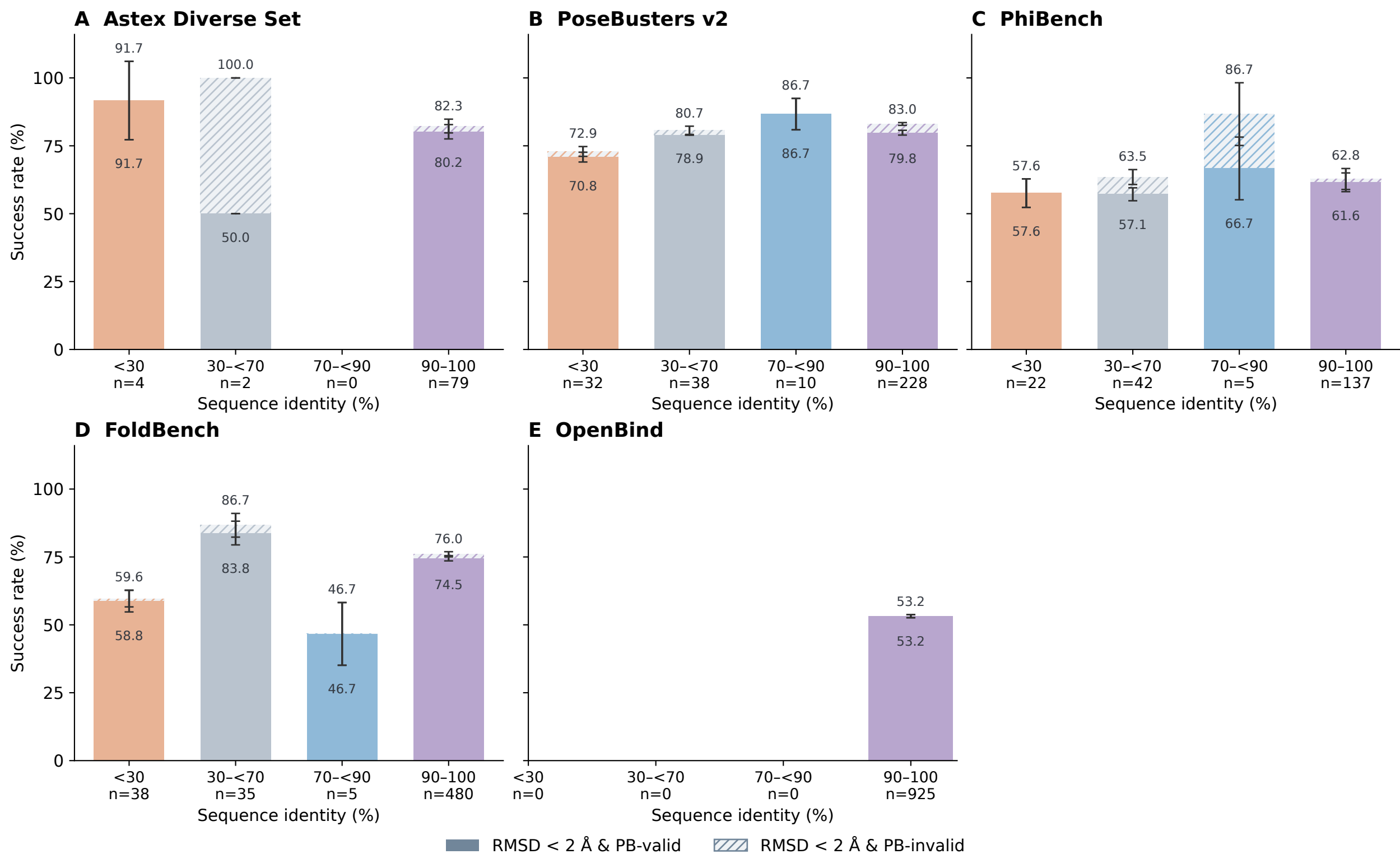
Sequence identity (%)

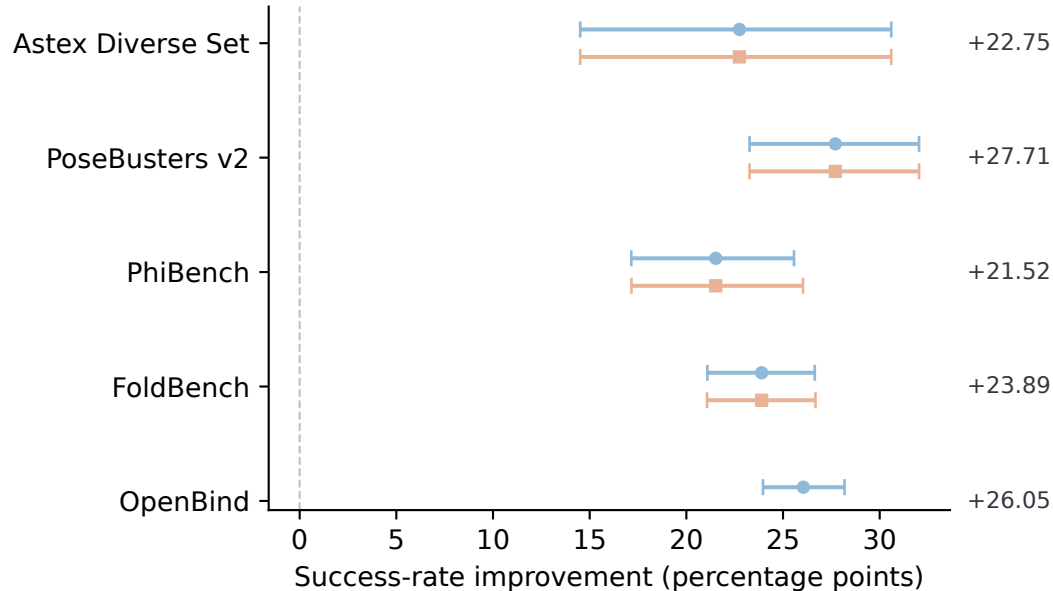
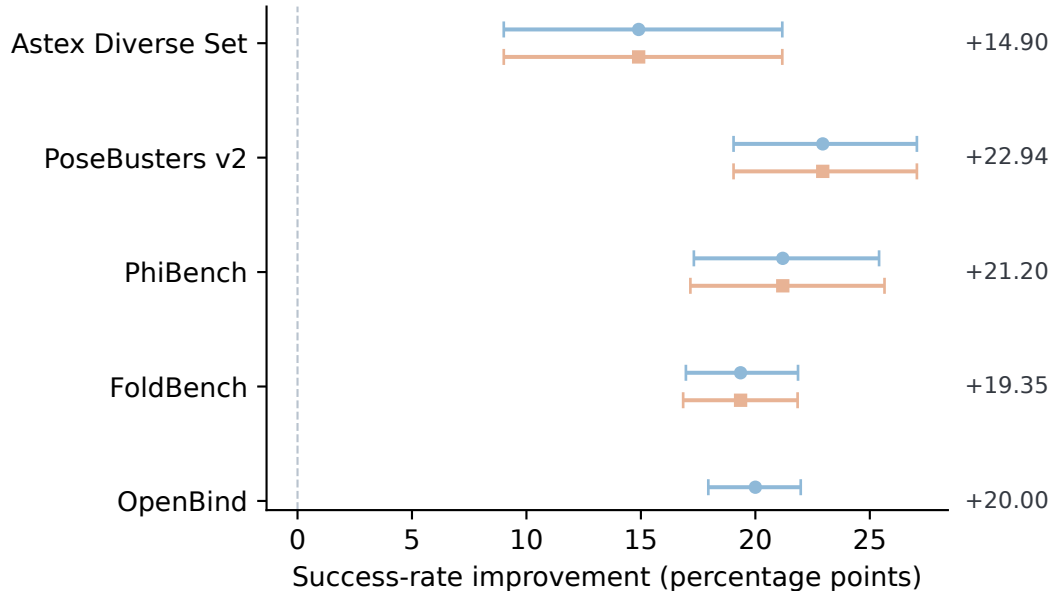
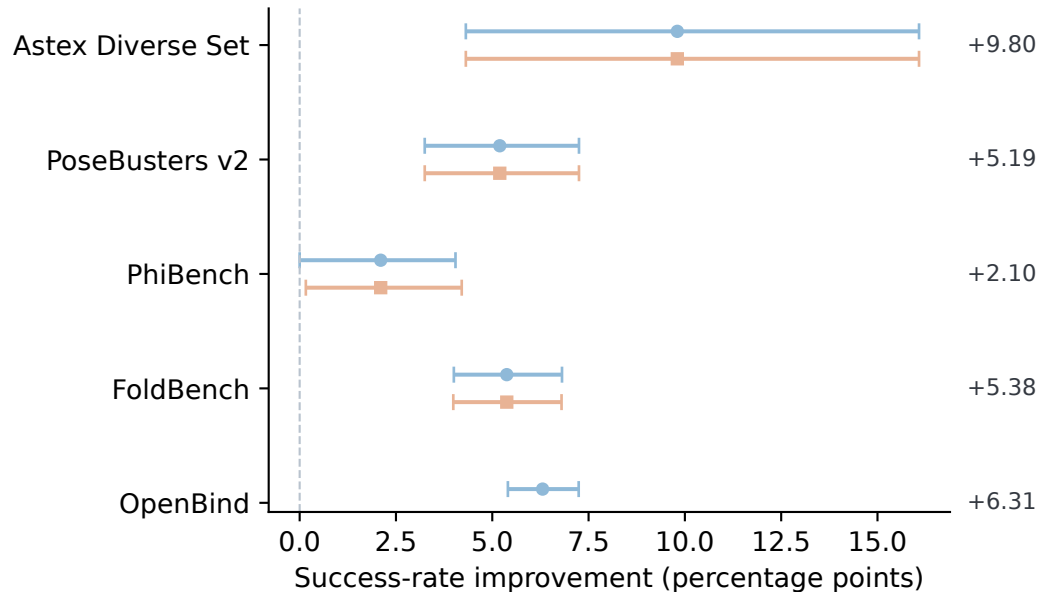
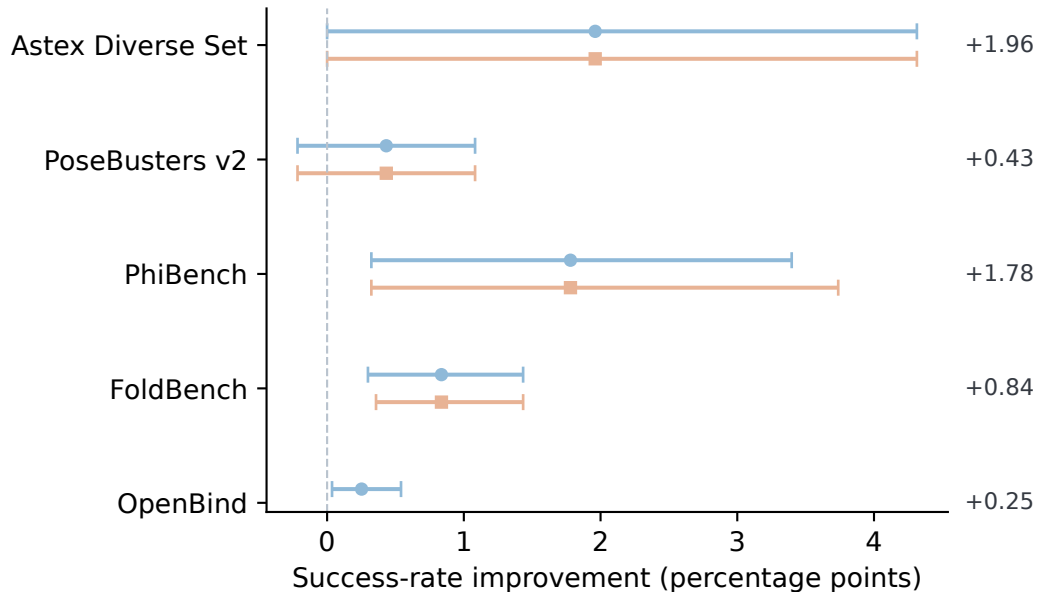
- <30%
- 30-70%
- 70-90%
- 90-100%

Ligand-protein overlap

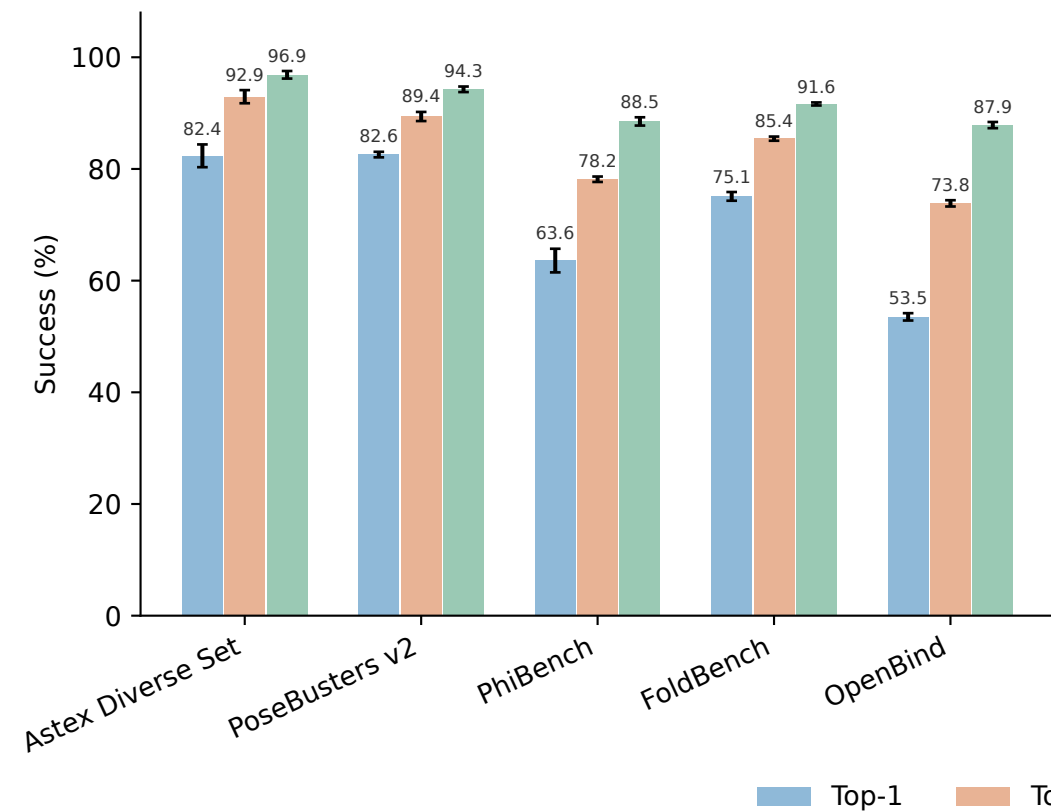
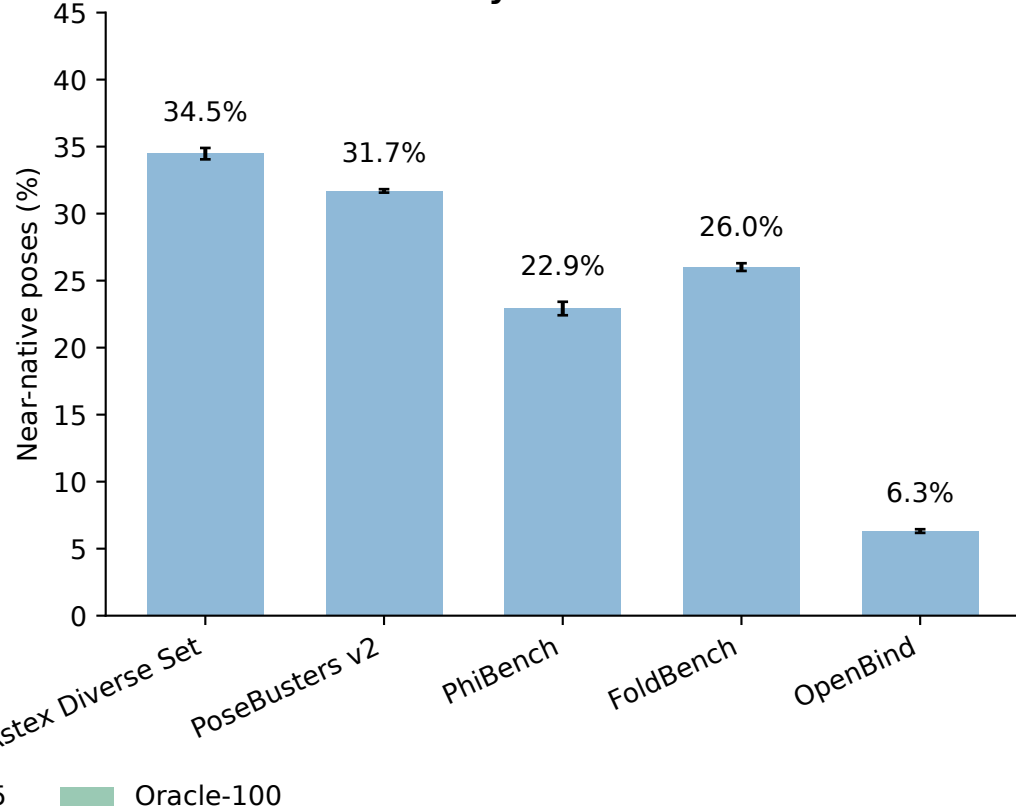
- Same-sample overlap
- Separate matches
- Ligand only
- Sequence only
- Neither observed

A Ligand similarity**B RMSD < 2 Å & PB-valid**

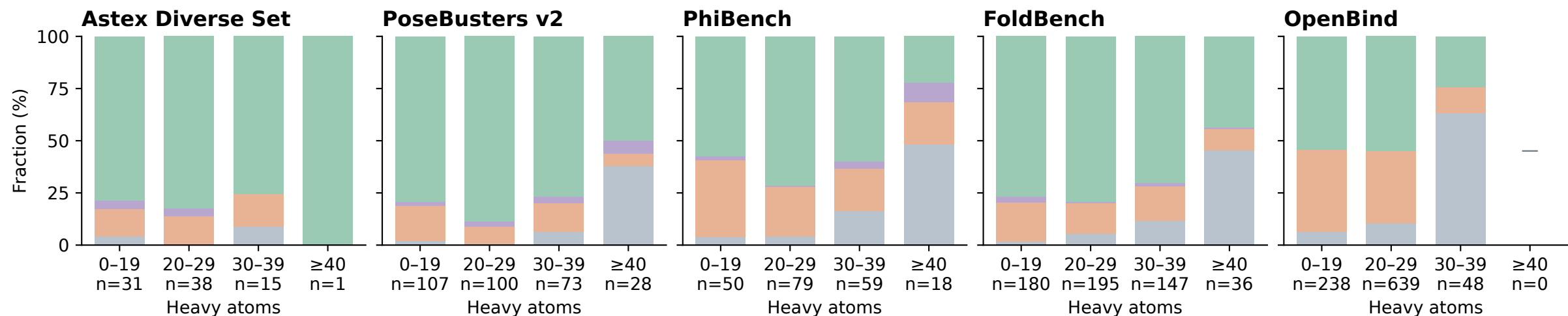


A Refined – raw, no chirality filter**B Refined – raw, with chirality filter****C Chirality filter off → on, raw****D Chirality filter off → on, refined**

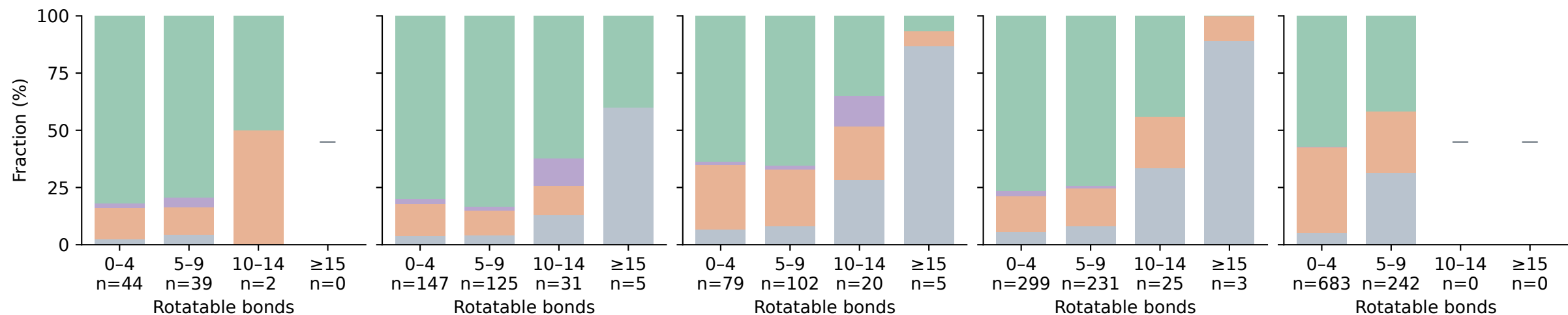
● Complex bootstrap ■ PDB-cluster bootstrap

A Selection**B Near-native density**

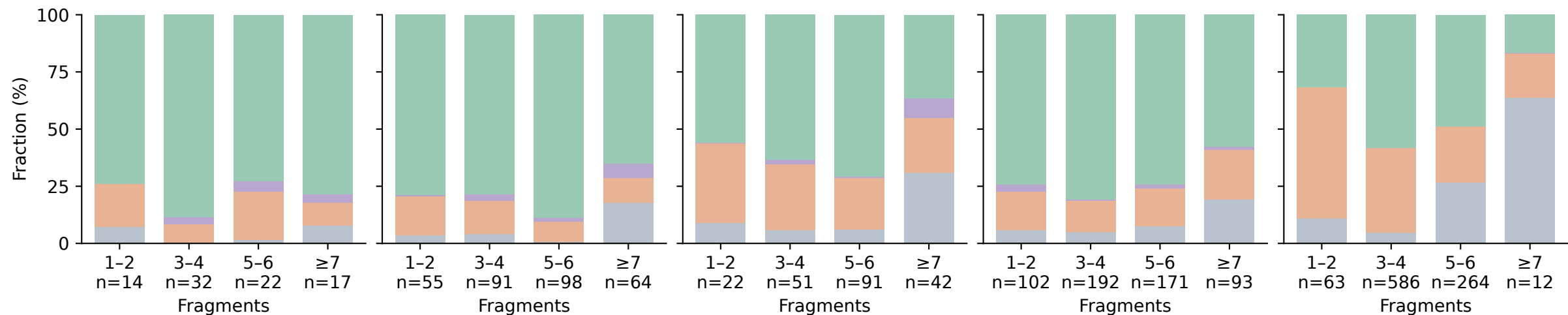
A Ligand size dependence



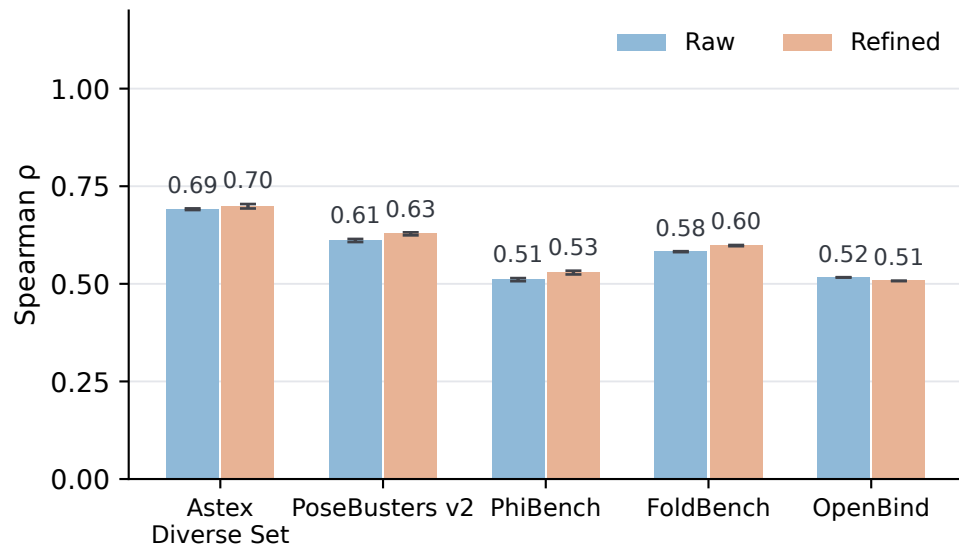
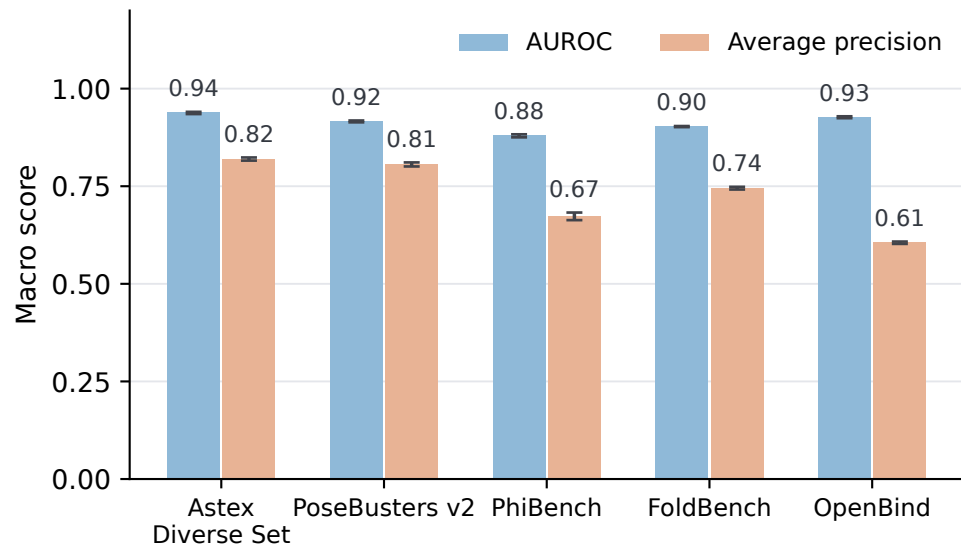
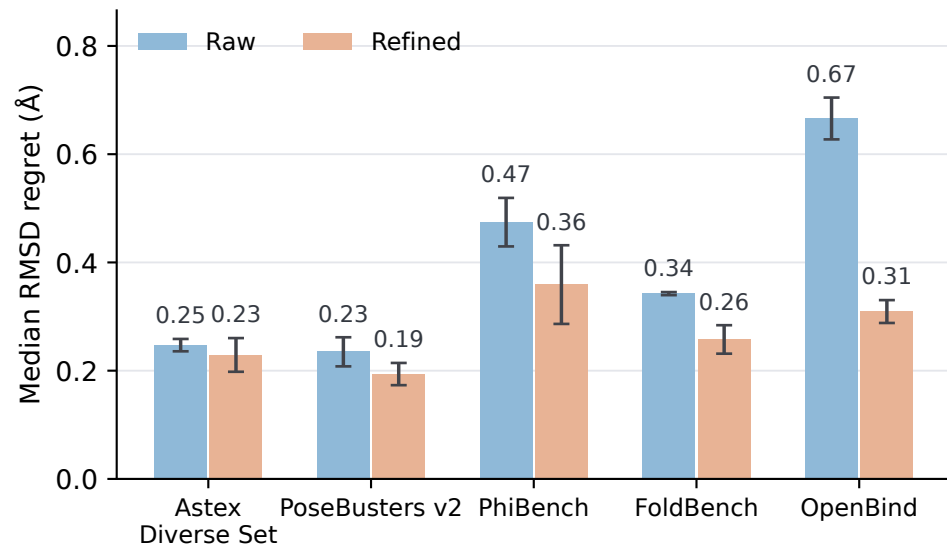
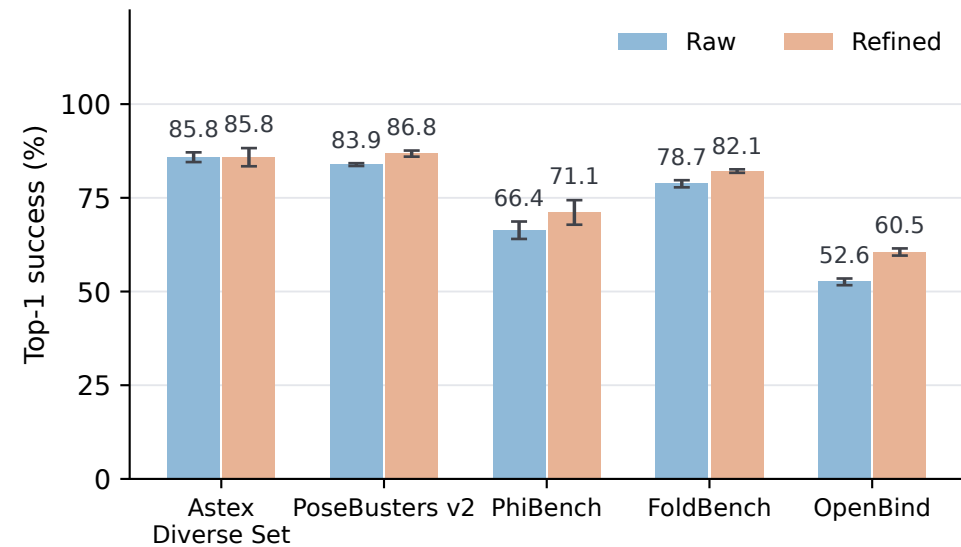
B Rotatable bond dependence

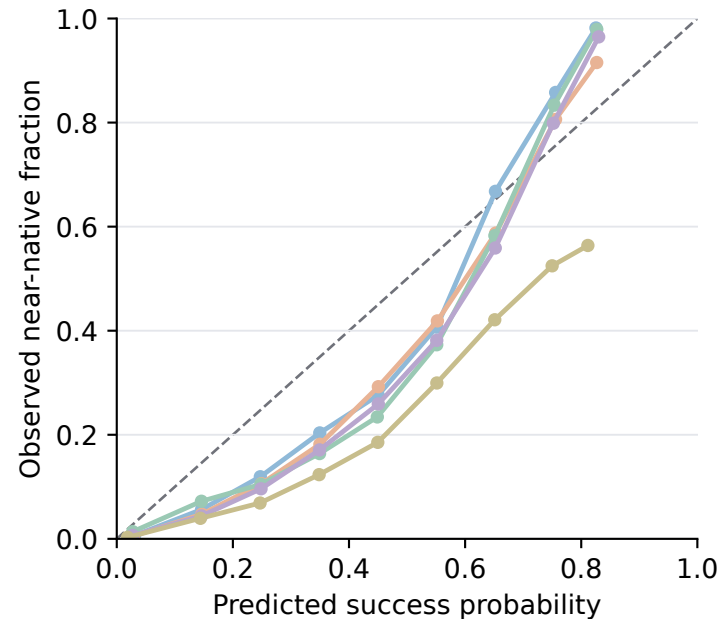
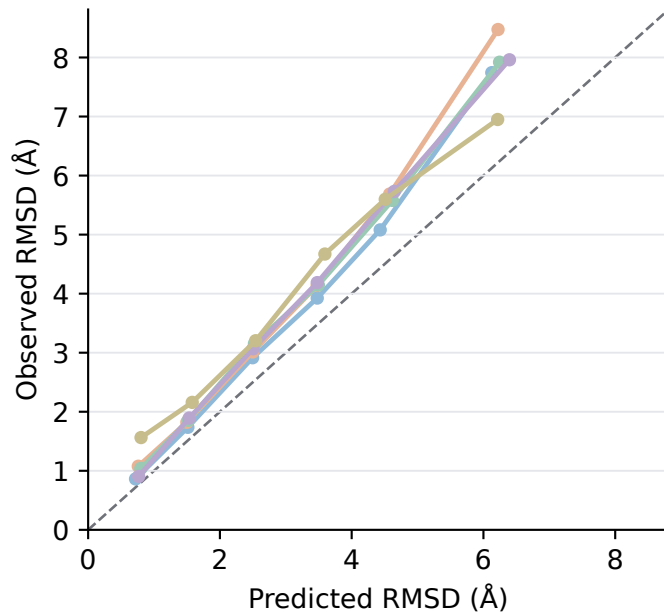
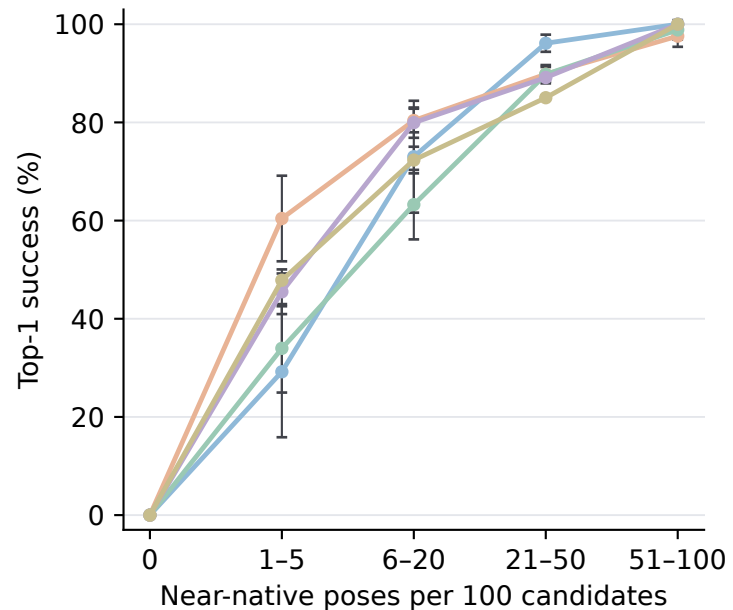


C Fragment count dependence



■ Near-native pose generation failure ■ Selection failure ■ PB failure ■ Success

A Within-bank ranking**B Near-native discrimination****C Selection regret****D Selection given a successful candidate**

A Success-probability reliability**B RMSD prediction reliability****C Near-native candidate density**

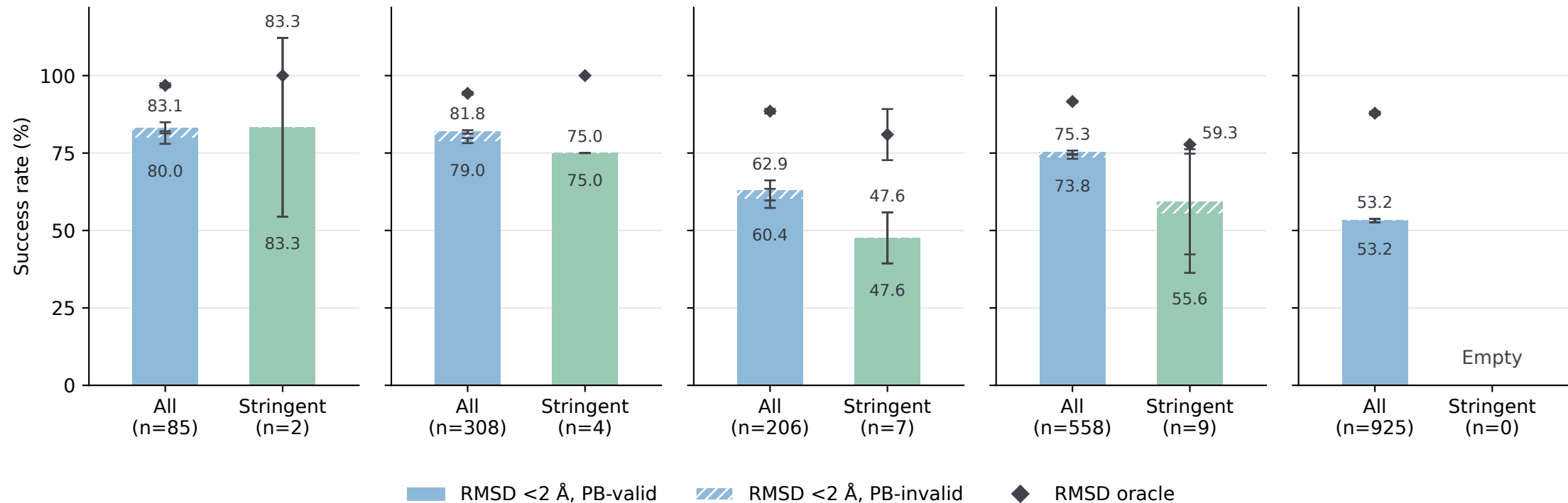
Astex Diverse Set

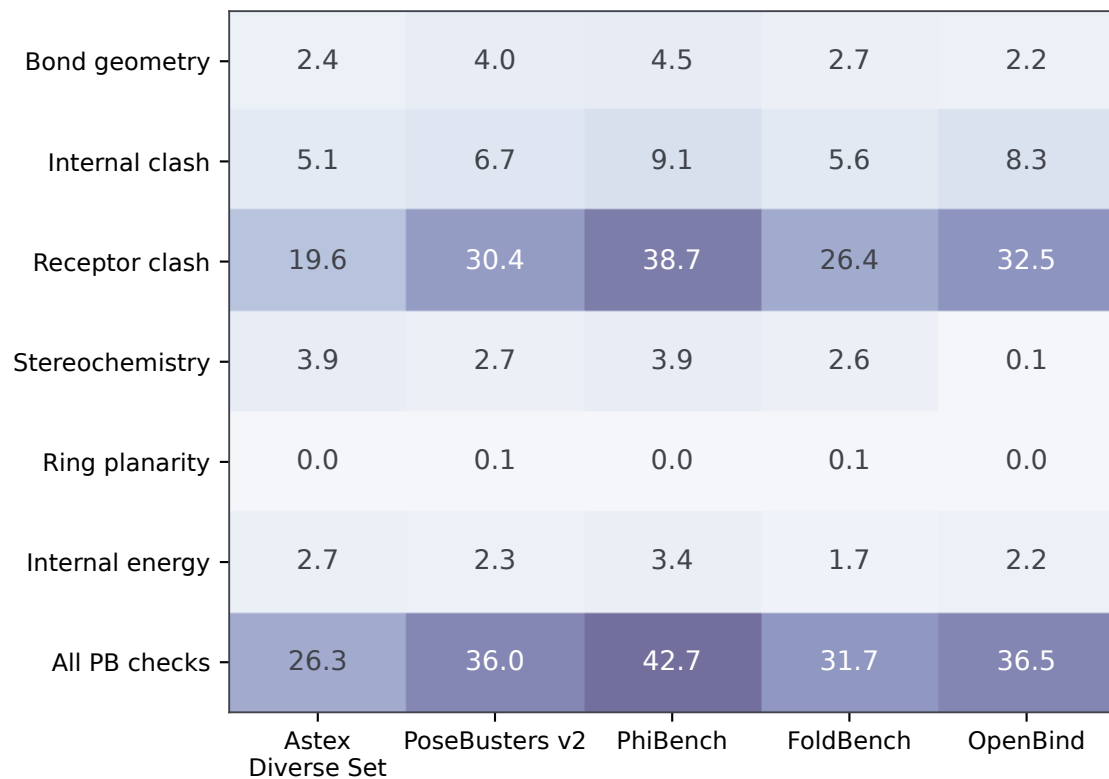
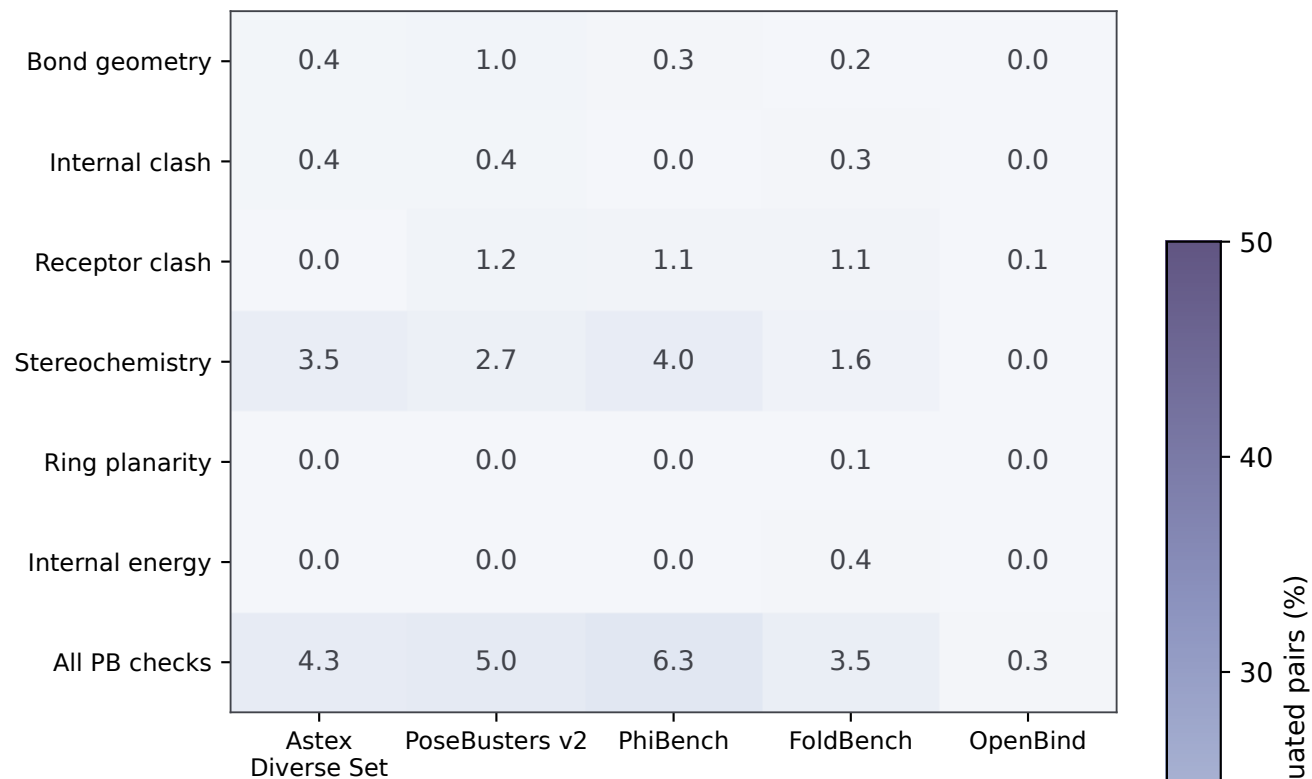
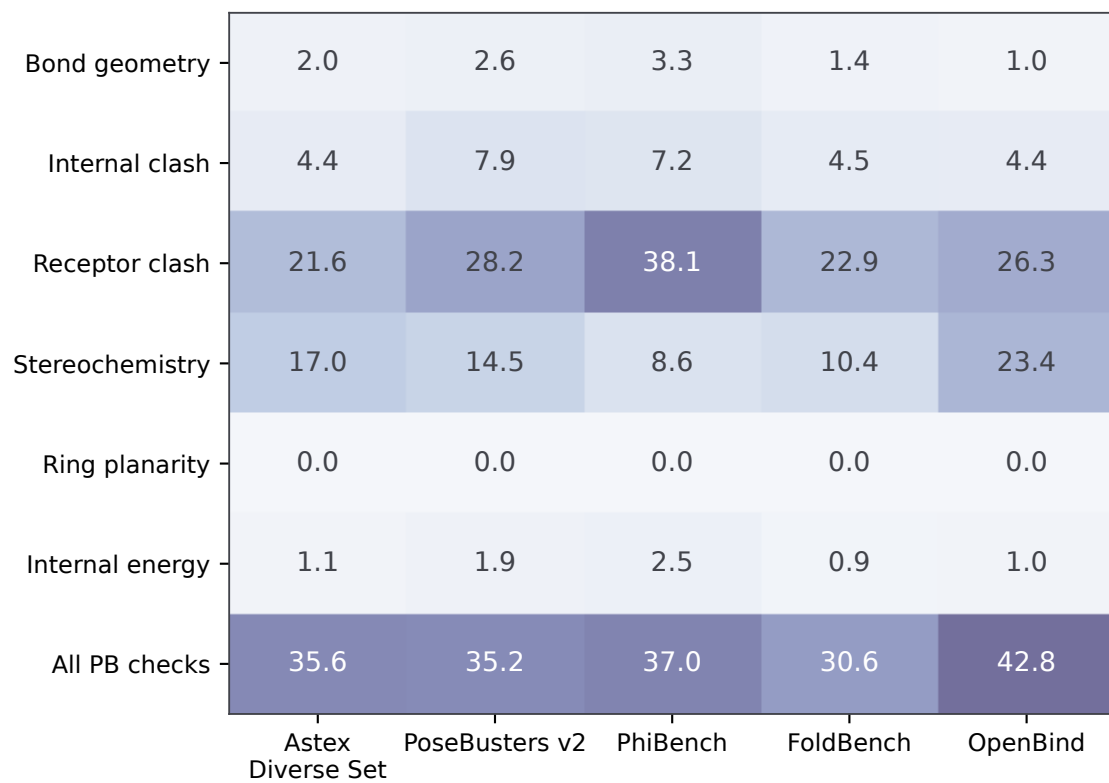
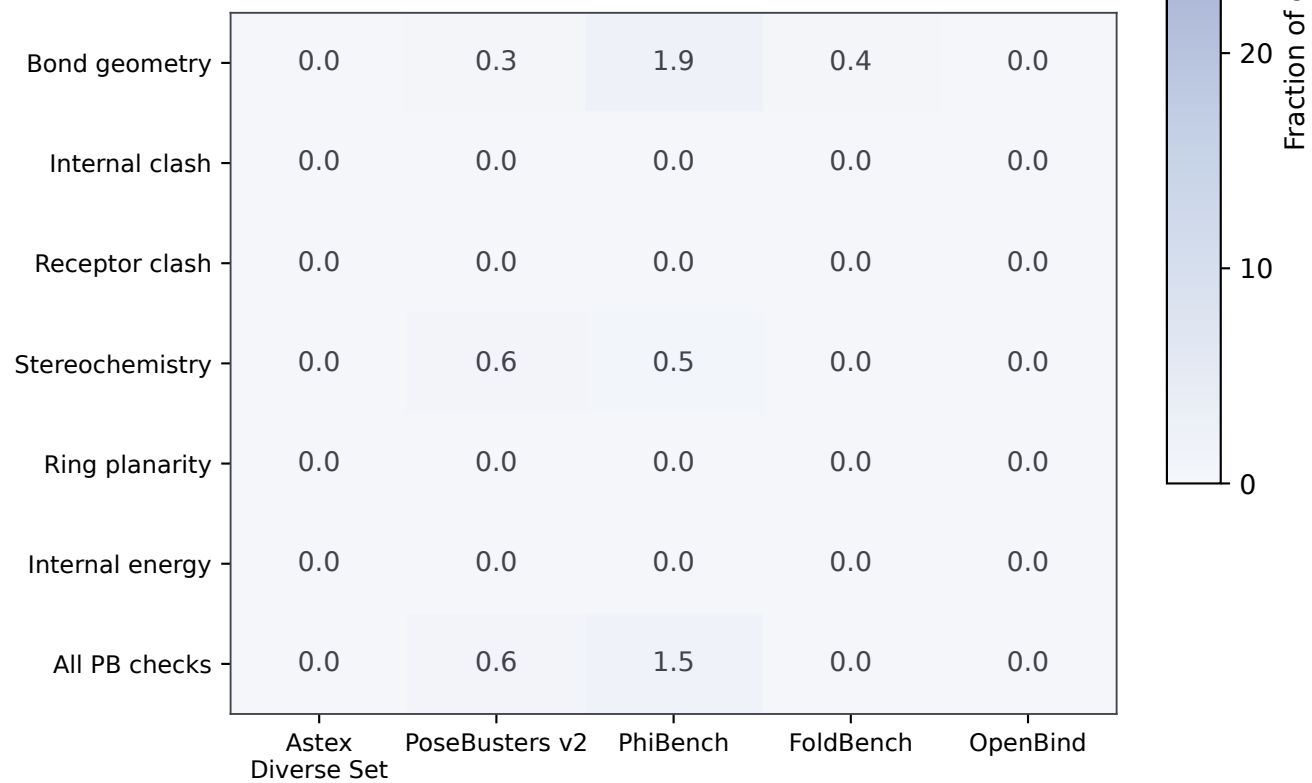
PoseBusters v2

PhiBench

FoldBench

OpenBind

A Astex Diverse Set**B PoseBusters v2****C PhiBench****D FoldBench****E OpenBind**

A Raw selected-pose failure**B Refined selected-pose failure****C Matched candidate: fail → pass****D Matched candidate: pass → fail**

Fraction of evaluated pairs (%)

50

40

30

20

10

0

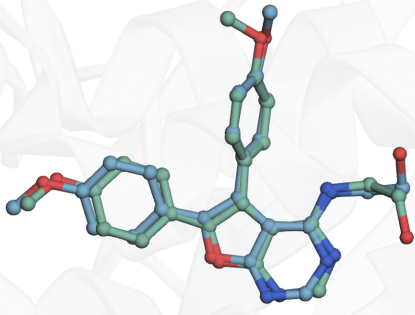
Successful pose

Refinement rescue

Selection failure

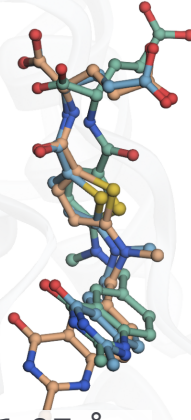
A Astex Diverse Set

2BR1



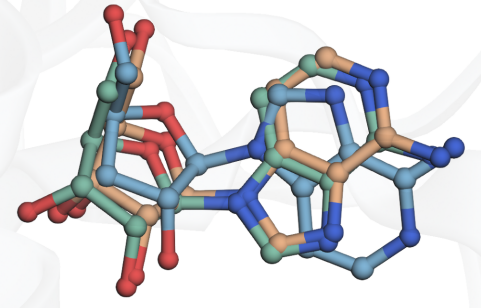
Selected RMSD: 0.76 Å

1HVY



Selected RMSD: 1.97 Å
Raw (same pose): 2.32 Å

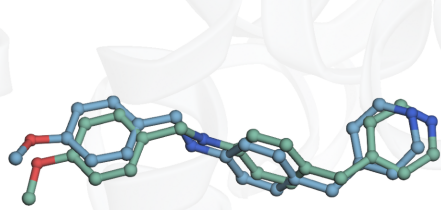
1HP0



Selected RMSD: 2.18 Å
Oracle RMSD: 0.51 Å

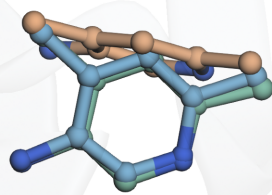
B PoseBusters v2

7P5T · 5YG



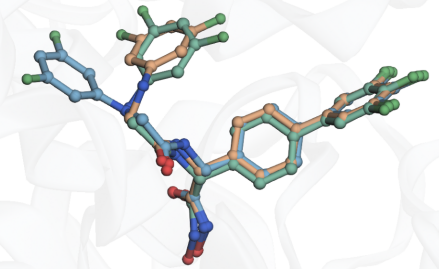
Selected RMSD: 0.88 Å

7FB7 · 8NF



Selected RMSD: 0.30 Å
Raw (same pose): 2.08 Å

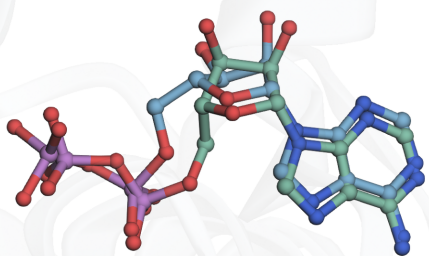
8EYE · X4I



Selected RMSD: 2.62 Å
Oracle RMSD: 0.66 Å

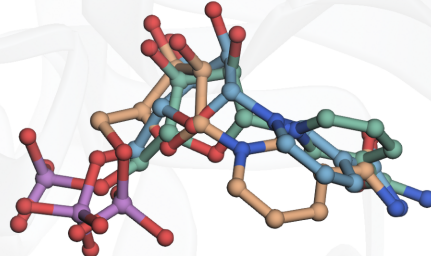
C PhiBench

9E8K · ADP · B · 1



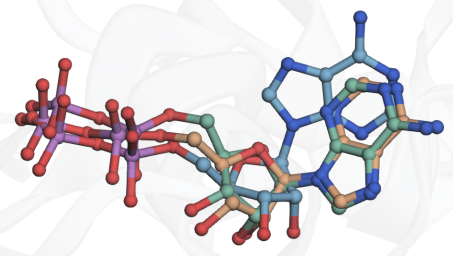
Selected RMSD: 1.03 Å

9FYB · NMN · A · 1 · B



Selected RMSD: 1.51 Å
Raw (same pose): 2.16 Å

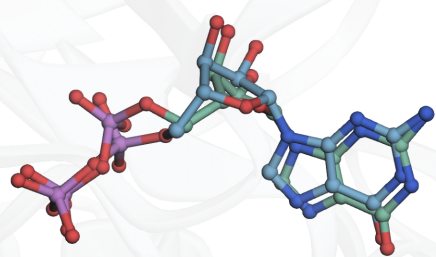
9E8N · ADP · D · 1 · E



Selected RMSD: 2.27 Å
Oracle RMSD: 0.64 Å

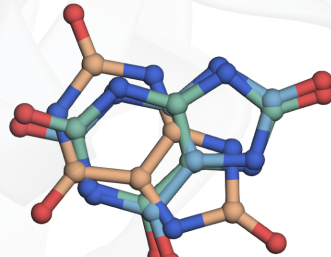
D FoldBench

7VF6 · GDP (E)



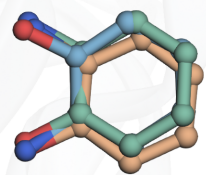
Selected RMSD: 0.93 Å

8BMV · URC (C)



Selected RMSD: 0.27 Å
Raw (same pose): 2.10 Å

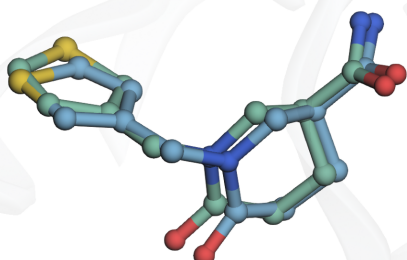
8IHG · 2AF (G)



Selected RMSD: 2.24 Å
Oracle RMSD: 0.60 Å

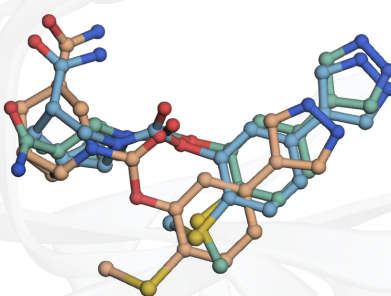
E OpenBind

A71EV2A-X5608B



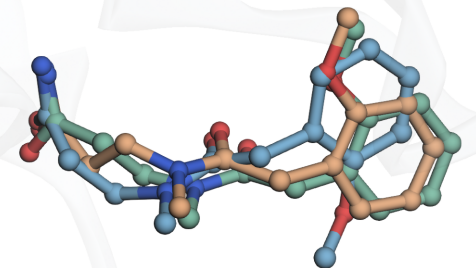
Selected RMSD: 1.12 Å

A71EV2A-X7379A



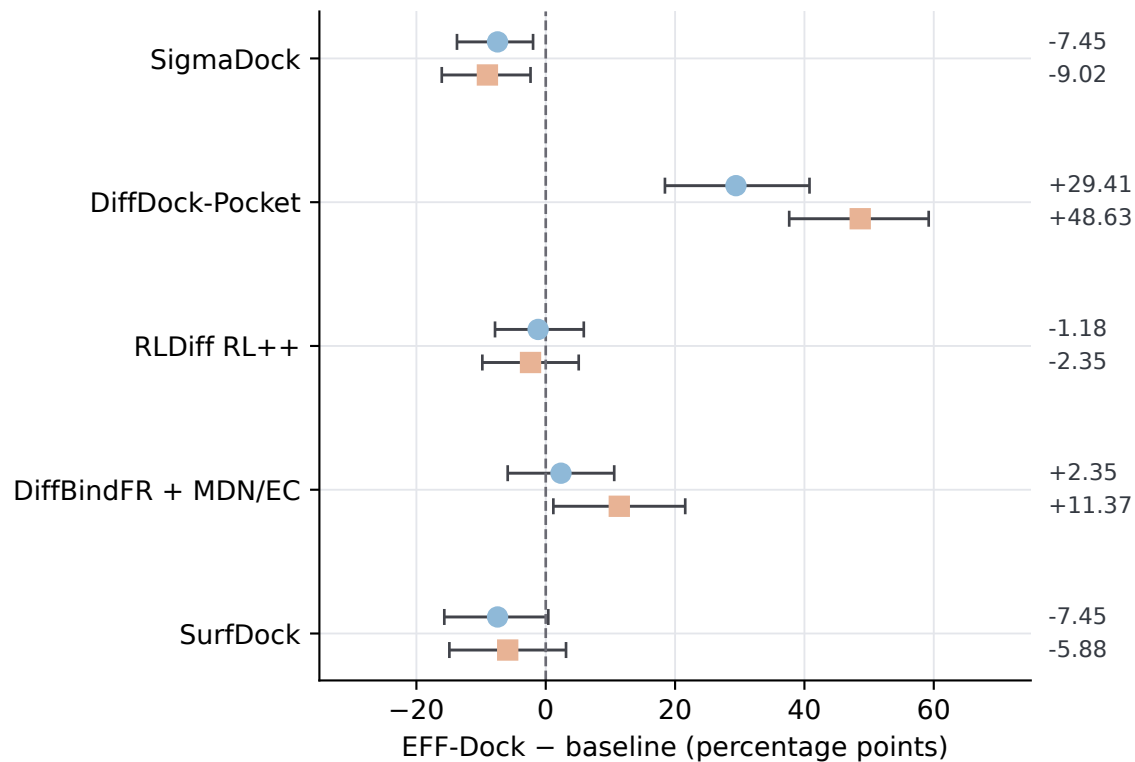
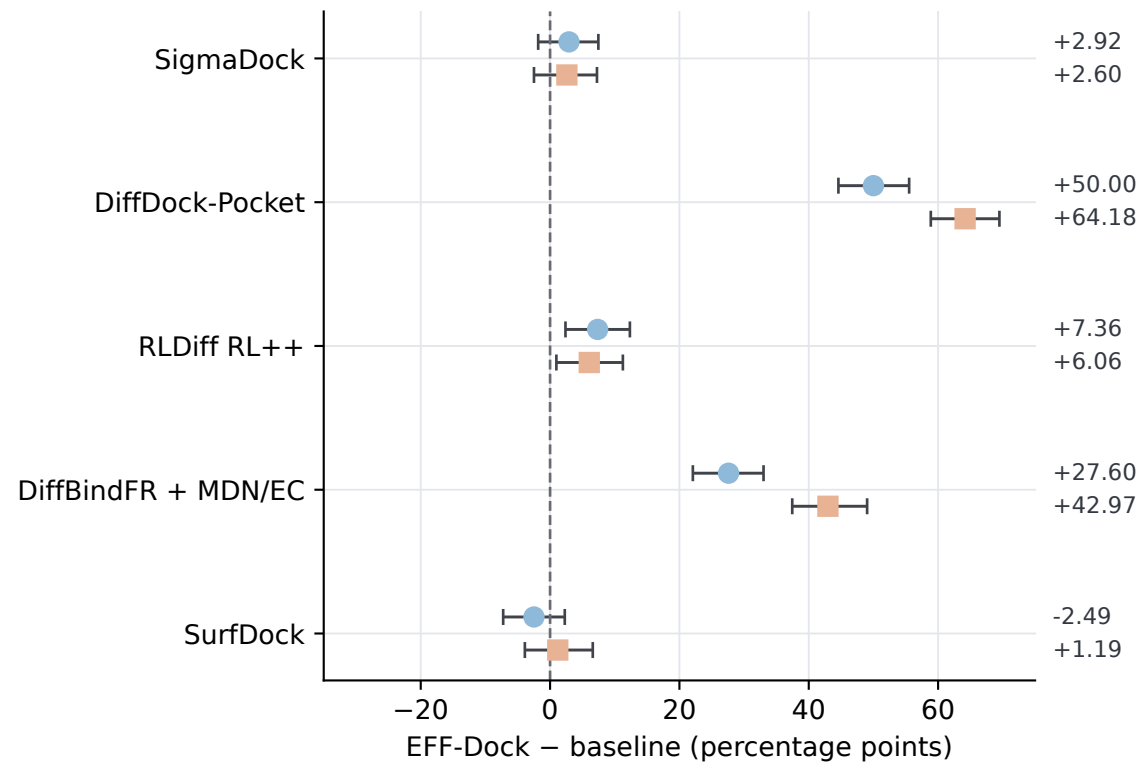
Selected RMSD: 1.60 Å
Raw (same pose): 2.87 Å

A71EV2A-X3977A



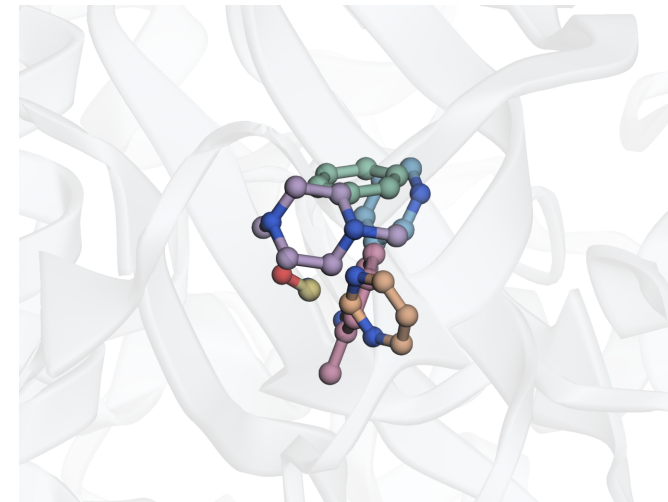
Selected RMSD: 2.29 Å
Oracle RMSD: 0.85 Å

Crystal carbon Selected carbon Raw / oracle carbon N O S P F

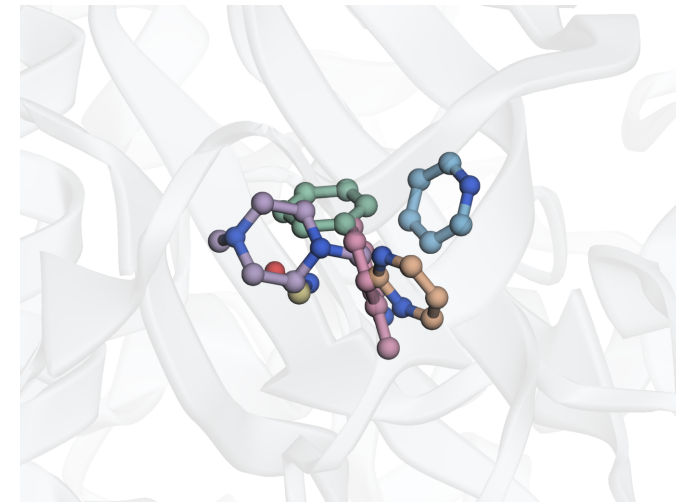
A Astex Diverse Set**B PoseBusters v2**

● RMSD success ■ PB-valid success

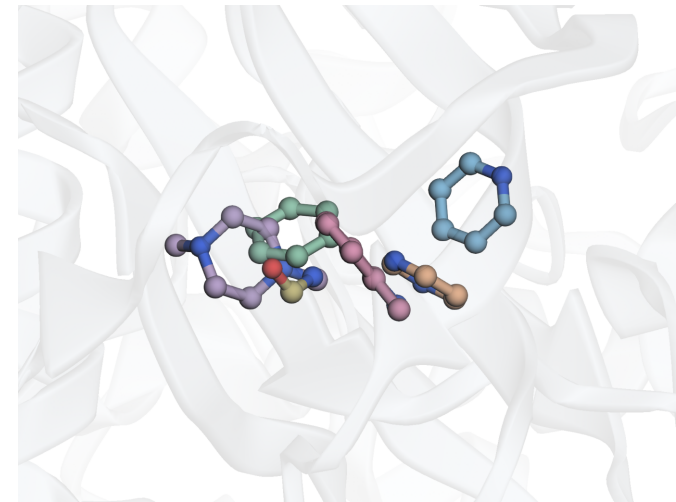
$t = 0.00$



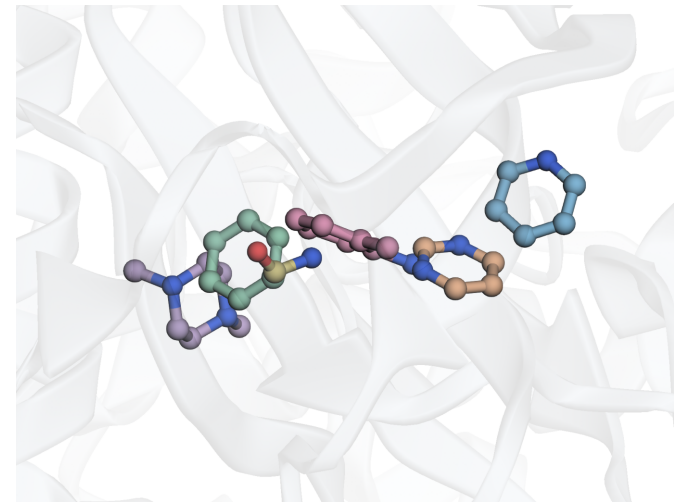
$t = 0.27$



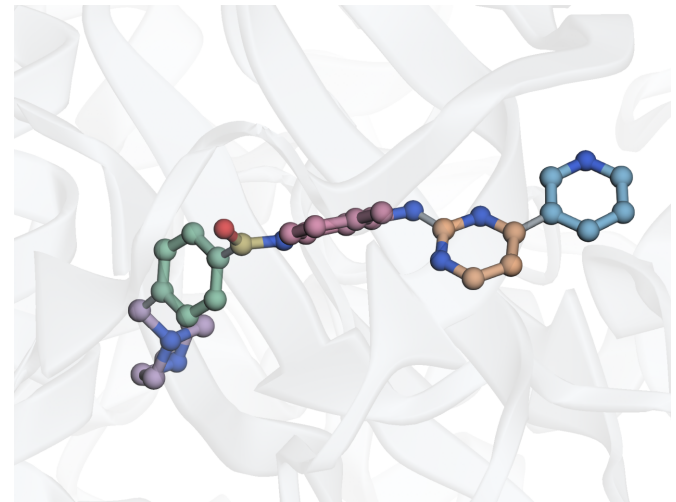
$t = 0.49$

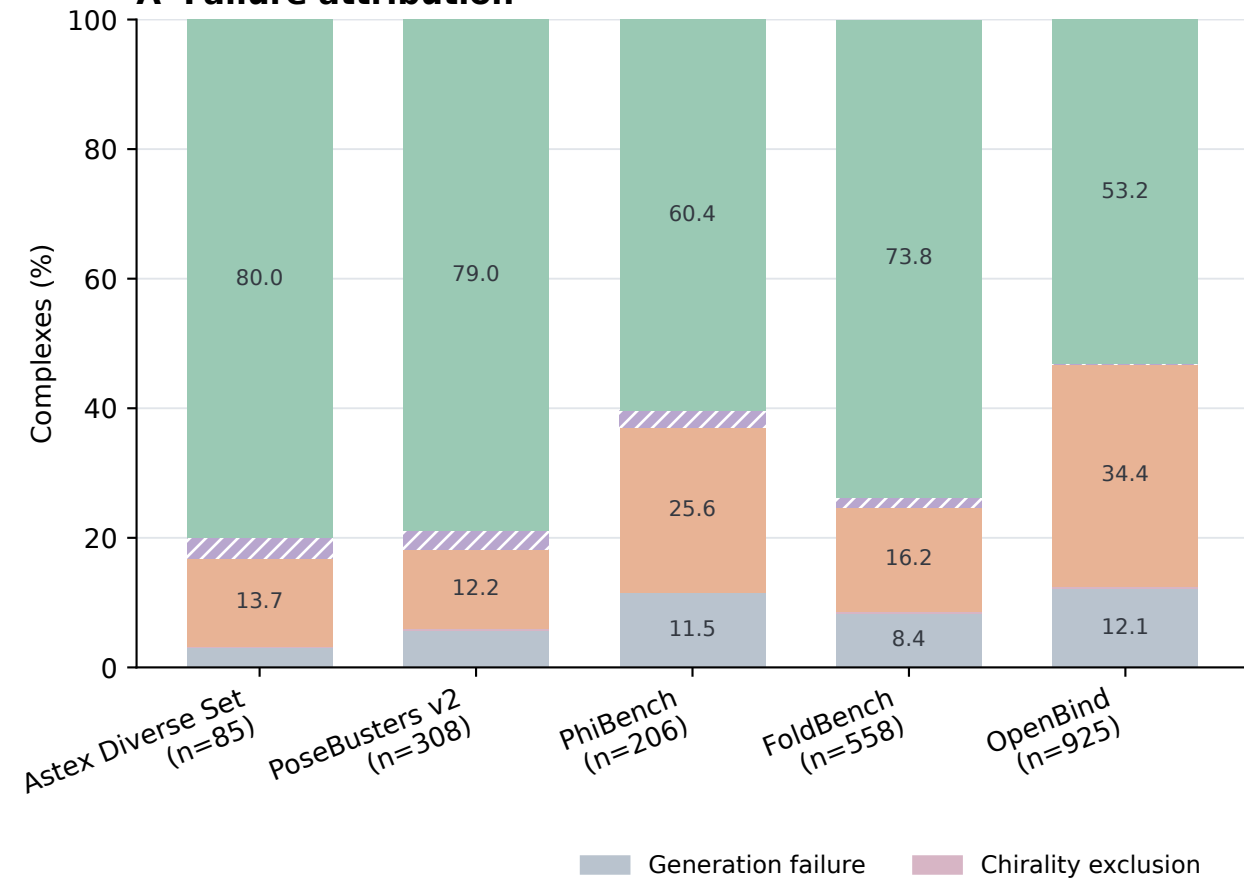


$t = 0.78$



$t = 1.00$



A Failure attribution**B Selection regret**